

RETAIL INDUSTRY MASTER DATA MANAGEMENT AND SERVICE-ORIENTED
ARCHITECTURE IMPLEMENTATION: BASIC ACTION RESEARCH

by

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RETAIL INDUSTRY MASTER DATA MANAGEMENT AND SERVICE-ORIENTED
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ABSTRACT

The retail industry is one of the sectors experiencing rapid growth over the past decade. This qualitative action research (AR) study explored the implementation of Master Data Management (MDM) and Service-Oriented Architecture (SOA) in the retail industry. The research examined the role of effective data management systems and event-driven architecture in improving business insights, including customer satisfaction and employee productivity. A narrative and grounded theory data analysis was used, collecting data from interviews with twenty business and technical experts. The findings suggest that MDM and SOA significantly enhance data quality across retail enterprises, helping improve decision-making processes. Challenges included difficulties in data governance and integrating information across departments. The study concludes that MDM and SOA frameworks are crucial for retail businesses to maintain a competitive advantage. Recommendations include further research into overcoming barriers to MDM and SOA implementation and exploring the impact on real-time data analysis.

DEDICATION

This dissertation is dedicated to those who have inspired, supported, and believed in me throughout this journey. To my family, for your unwavering love and encouragement. Your faith in me has been the foundation of my success and has motivated me to persevere through challenges. To my mentors and advisors, for your guidance and wisdom. Your insights and expertise have been instrumental in shaping this work. To my friends and colleagues, for your support and camaraderie. Your encouragement and understanding have made this journey more meaningful. To my daughter Sagie bear, you are the reason why I continue to push boundaries and push myself to be the best at everything I do so you understand that “WE” can do anything we want. To my Dr. DeAnna Bond, thank you for encouraging me and pushing me through this process and believing in me when I didn't believe that this was my purpose and journey. And to all who pursue knowledge and growth, may this work serve as a testament to the power of determination, curiosity, and perseverance. Thank you all for being a part of this milestone in my life.

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Chapter 1

Introduction

The goal of this research was to investigate how master data management practices, as understood by leadership teams, bridge the gap between technological advancements and business outcomes in the retail industry. Retail organizations rely on effective data strategies to support decision-making and reporting (Arthofer & Girardi, 2017). Since the 1980s, data management has played a critical role in driving enterprise transformation through effective knowledge use (Legner et al., 2020). However, academic research is limited to the impact of these systems on business performance (De Camargo Fiorini et al., 2018). Leadership perspectives will provide valuable context for understanding the implementation of data systems to meet organizational goals and overcome challenges. By gathering qualitative insights from retail leaders, the study explores how data management practices support organizational goals and address challenges, contributing to a deeper understanding of how these strategies improve decision-making and performance in the retail sector.

Chapter 1 summarized the background of the problem, the significance of the study, population and sample, nature of the study, methodology, design, research questions, conceptual framework, diffusion of innovation theory, resource-based theory, dynamic capabilities theory, definitions and assumptions, limitations, and delimitations. Chapter 1 included a foundation for how the study was conducted using a sample population of technology professionals and methodology used to capture the research data.

Background of the Problem

Effective data management presents substantial challenges within the retail industry due to the volume and sensitivity of customer data collected. Participants in prior studies have emphasized that much of this data includes personal customer information and requires the need for data confidentiality (Smith, 2020). Additionally, qualitative accounts suggest that the adoption of Internet of Things (IoT) technologies for data collection has heightened awareness of potential customer data security vulnerabilities. Retailers rely heavily on customer insights to predict demand and ensure products and services meet customer needs and preferences. However, challenges arise in interpreting and analyzing this data, as respondents have reported misalignments within enterprise data initiatives. According to Johnson (2021), challenges are particularly pronounced in context where collaboration and integration are required but hindered by the persistence of isolated activities and reflect a lack of coordinated data management strategies. Addressing these complexities calls for an exploration of integrated practices that reflect participants' experiences and align with organization objectives to leverage customer insights effectively.

Retail companies face interconnected challenges in data management that undermine informed decision-making and operational efficiency. Ying et al. (2020) highlighted that retailers struggle with issues in data management, particularly in consolidating information from large sets and complex data structures such as customer, supplier, and vendor data. Johnson et al. (2024) highlights that when retailers struggle to combine data effectively, this can hinder the correlation and mapping data to respective stakeholders. This can result in inaccurate data insights which leads to poor decision-

making. Additionally, data duplication and redundancy from inefficient data management practices complicate the process. The complexities of extraction, movement, and combination of data compound these challenges. A great amount of time is wasted on the manual process of gathering, verifying, and consolidating, and formatting which reduces time for strategic planning. These inefficiencies are also linked to poor forecasting, which can harm organizational performance. The inability to handle large volumes of data effectively contributes to incorrect business predictions and suboptimal decisions. To address these data management challenges, the adoption of Master Data Management (MDM) and Service-Oriented Architecture (SOA) is essential.

Expanding data management capabilities is essential for organizations to enhance their portfolios through effective implementation of data management strategies. Ying et al. (2020) emphasized that advancements in data management foster automation and the development of Big Data Management (BDM), which is critical for analyzing and managing vast volumes of data. Retailers are increasingly focused on using BDM to address challenges such as understanding customer needs and preferences, However, there are inconsistencies in defining scope, purpose, and role of BDM hinders its full potential which limits its effectiveness in retail data management (Ying et al., 2020). Therefore, businesses must adopt standardized approaches to BDM to optimize data management and improve overall business performance.

The increase in the number of network devices and the involvement of data in decision-making in the retail industry creates a need for a reliable data management framework. Cohen et al. (2017) argued identifying, categorizing, and assessing the quality of Big Data is a challenge for many retailers. According to Cohen et al. (2017),

retailers have adopted the practice of using data management in predictive analysis to understand changes in demand thus improving strategic planning. Therefore, retailers focus on storing data on customer purchases, spending habits, and preferences.

The rapid growth of data volume and complexity of information management have heightened concerns about data security among information technology (IT) security professionals and business consultants. Many organizations have suffered financial losses due to data breaches, prompting a shift toward adopting Master Data Management (Doe & Smith, 2022). The global MDM market is projected to grow from United States Dollar (USD) 6 billion in 2019 to over USD 20 billion by 2027, reflecting the broader adoption of data management solutions across various business sectors (Zhao & Lee, 2021). This trend underscores the need for robust data solutions, as businesses in retail and manufacturing rely on consistency and automated data processes to maintain operational efficiency (Petersilge, 2020). Adopting MDM solutions is critical for organizations seeking to mitigate data security risks and enhance performance in competitive markets.

The growing complexity of retail data requires the industry to adapt its approaches to meet evolving customer needs. Matsunaga (2012) highlighted the increasing variety, velocity, and value of retail data, necessitating new strategies to match shifting customer models. While consumers favor e-commerce, offering a personalized experience remains crucial (Chandramana, 2023). Additionally, the retail sector's emphasis on understanding customer preferences has driven the adoption of data collection and analytics (Gan & Dai, 2014). Despite its potential, data from diverse channels poses challenges for informed decision-making, primarily due to unreliable data

management tools and misaligned organizational goals (Chandramana, 2023). These challenges underscore the need for precise data interpretation and alignment with strategic objectives. To remain competitive, retailers must invest in advanced data management solutions that enable actionable insights and support personalized customer engagement.

Accurate business data is essential for organizations to develop effective business plans and make informed decisions. Master Data Management (MDM) supports business growth and new revenue generation by offering a structured framework for managing data across various data domains, such as business operations, employee human resource data, and customer data (Johnson et al., 2024). The hierarchical organization of data enhances visibility and maximizes operational efficiency, enabling businesses to leverage MDM to create value (Lee & Patel, 2022). Implementing MDM ensures that organizations can optimize their data resources, fostering sustainable growth and competitive advantages.

Effective data management is essential for gaining a competitive edge in the retail industry. Brajesh (2016) emphasized that conventional data management strategies face challenges such as multiple data touchpoints, which limit effectiveness. Annansingh and Sesay (2022) stated the importance of a unified view of data, including inventory, customer orders, and shipping, alongside regular updates across platforms and systems. Irrelevant data or data siloed systems in traditional data management approaches can have long-term negative impacts on the retail operations, reducing efficiency and competitiveness (Brajesh, 2016). To remain competitive, retailers must adopt integrated

data management strategies that eliminate silos, ensure data relevance, and provide a holistic view of operations.

Implementing Service-Oriented Architecture (SOA) and Master Data Management (MDM) is crucial for enhancing data management practices in the retail industry. Redundant data points remain a major issue, as multiple data warehouses and repositories, often updated manually, lead to inconsistencies and inefficiencies (Annansingh & Sesay, 2022; Özdemir & Aydin, 2020). Additionally, reliance on the Internet of Things (IoT) increases data exposure to third parties, creating security risks for sensitive customer and organizational data (Brajesh, 2016). These challenges undermine retail operations, resulting in data inconsistencies, compromised strategies, and potential financial losses. Overwhelming traditional practices, such as manual updates and redundant repositories, exacerbate inefficiencies, while IoT driven vulnerabilities heighten the risk of data breaches. To enhance operational efficiency and ensure data security, retailers must adopt secure, unified data management systems that mitigate redundancies and protect sensitive information.

Retail companies face significant challenges in maintaining large data structures and ensuring effective data management. Chandra et al. (2022) identified challenges such as data inconsistency, silos, duplication, quality concerns, complex integration, limited analytical capabilities, all which undermine the effectiveness of data analysis and reporting tools. While Big Data Management offers a potential solution, it also introduces additional complexities for retailers (Cauhan et al., 2017; Ying et al., 2020). These challenges hinder retailers' ability to derive actionable insights and align data management with organizational goals. Although Big Data Management can improve

tool reliability and data utility, its implementation requires overcoming technical and operational barriers. To enhance data management effectiveness, retailers must address structural inefficiencies while carefully integrating Big Data solutions to optimize data reliability and insight generation.

Predictive analytics and modern data management strategies can address critical challenges in understanding customer needs and handling data storage in retail. Cohen et al. (2017) recommended predictive analytics for gaining deeper insights into customer behaviors and evolving needs. Smith et al., (2019) identifies data storage as a primary challenge and advocates for Master Data Management to streamline and centralize data handling. However, conventional data management strategies often face issues such as multiple touch points, while modern approaches, like Internet of Things (IoT), introduce risks to data security (Annansingh & Sesay, 2022). These challenges highlight the need for advanced, integrated approaches to data management. Predictive analytics enhances customer understanding, while centralized solutions like Master Data Management reduce inefficiencies. However, the adoption of IoT requires robust security measures to protect sensitive customer information. To optimize data management, retailers must combine predictive analytics, centralized data systems, and enhanced security protocols to address storage inefficiencies and protect customer data.

The implementation of Service-Oriented Architecture (SOA) and Master Data Management (MDM) is essential in leveraging customer data to drive process improvements and increase sales in the retail industry. Brajesh (2016) emphasized that customer data is one of the retail industry's most valuable assets, necessitating effective tools to manage duplicates, inconsistencies, and inaccuracies. The issue concerns the

discrepancies in product data that SOA and MDM can address, which could otherwise result in customer attrition and competitive disadvantage. By effectively managing customer and product data through SOA and MDM, retailers can enhance their ability to identify improvements areas, ensure data accuracy and monitor growth, which are critical to maintaining high customer satisfaction and staying competitive in the market.

Problem Statement

Retail organizations experience significant challenges in managing vast amounts of customers, vendors, and operational data due to increasing digital complexity and evolving consumer expectations. According to De Vass et al. (2021), the retail industry collects a large volume of customer data that must protect sensitive personal information through effective data management tools. The increasing use of Internet of Things (IoT) technologies for data collection has heightened security threats and complicated data management processes. Retailers struggle with consolidating and integrating data from sources like customers, suppliers, and vendors resulting in data duplication, redundancy, and inaccuracies. Therefore, it is essential to explore how the implementation of Master Data Management (MDM) and Service-Oriented Architecture (SOA) can intervene to improve data quality in the retail industry. Essentially, this can ensure the need to optimize overall data management and operational efficiency.

Purpose of the Study

The purpose of this qualitative basic action research (AR) study was to evaluate the role of Master Data Management (MDM) and Service Oriented Architecture (SOA) in improving data quality and integration within the retail industry. Assessing how the implementation of Master Data Management (MDM) and Service-Oriented Architecture

(SOA) enhances the quality and integration of data across enterprises within the retail industry, particularly from the perspective of IT technicians and business analysts (Kraus et al., 2018). The following three objectives inform the purpose of the study:

1. To identify the key challenges and success factors in implementing MDM and SOA in the retail context.
2. To evaluate how MDM and SOA can be leveraged to improve data accuracy.
3. To develop recommendations for retail organizations and practitioners seeking to align data management practices with business objectives.

Achieving the three objectives could inform important decisions by various stakeholders. The research findings from the study could aid information technology enthusiasts in creating effective enterprise data strategies. Retail traders could rely on the data findings to make investment decisions that could enhance retail business efficiency. Lastly, entrepreneurs and/or consultants could rely on the findings to support organizational agility.

Population and Sample

Research studies rely on a large population and a sample of the large population to derive findings. According to Sudha (2017), a population refers to people, objects, or events with a specific characteristic and interest the researcher. These groups are defined by research objectives. A sample is a subset of the population chosen for the study. It is always smaller than the study population and can be used to make inferences about a larger group (Banerjee & Chaudhury, 2010). Appropriate sampling techniques are important to ensure that the sample accurately reflects the characteristics of the general population and to enable researchers to generalize their findings. The study ensured

trustworthiness through rigorous qualitative methods, focusing on credibility, transferability, and dependability in the data collection and analysis. Constraints such as time, cost, and availability often prevent researchers from studying the entire population, thus increasing the reliance on samples, which, if properly chosen, can provide insights that represent the population (Sudha, 2017). This section discussed the population and sample used in this study.

A global organization with over 300,000 employees and 15,000 eligible individuals agreed to participate in this basic action research study. The number of 20 participants, self-selected to participate. This AR study focused on the expert level of the study population based on the years they have worked in data management positions in the retail industry and their experience in implementing MDM and SOA. Business analysts and technical experts with more than five years of experience in data management in the retail industry and those who have implemented Master Data Management (MDM) and Service-Oriented Architecture (SOA) for more than five years were selected to participate in the research. This pool of potential participants brought an informative approach to the study as it related to the qualitative methodology with an action research design. The delimited population size was approximately 3000. The delimited population of 3000 was determined based on the total number of eligible participants from a large global organization with over 15,000 employee and self-selected individuals meeting the criteria of five years' experience in MDM and SOA implementation.

The sampling criteria was based on company categorization and encompassed practitioners who have worked on MDM Implementations of retail companies of large-scale enterprises. The criteria included:

- large-scale retail corporations: Encompassing multinational retailers with extensive operations (100+ locations in the US).
- diverse product offerings (2000+ products)
- wide customer base (2 million customers)
- apparel and fashion retailers: MDM and SOA implementations for managing product catalogs, inventory, and customer preferences.
- e-commerce Platforms: MDM and SOA implementations used to execute online retail operations, digital marketing strategies, and omni-channel customer engagement initiatives.

As it relates to the business functions, the sampling approach identified various business functions within retail organizations to assess the multifaceted impact of MDM and SOA. This involved engaging practitioners, technical architects, and analysts with direct experience with MDM implementations in the retail sector. Specifically, the study included practitioners from these departments within retail: sales/ marketing and operations/supply chain management.

A large and targeted sample size was essential in action research to collect comprehensive and meaningful data for sound conclusions. Lynch and Welch (2014) emphasized the importance of using a large sample size to ensure a broad range of data that facilitates reliable comparisons and conclusions. In this study, a total of 20 business architects and technical experts were recruited using purposive sampling, a method that

enabled researchers to select participants most likely to provide relevant and insight data (Etikan et al., 2016). Purposive sampling was particularly valuable in this study focusing on specific characteristics or phenomena where random sampling may fail to capture the necessary depth of information. By targeting a specific group of experts, this approach ensured that collection of detailed and accurate insights, enhancing the study's depth and validity. The use of purposive sampling in this research ensured that the data gathered is both relevant and robust, contributing to the formulation of sound and actionable conclusions.

Significance of the Study

The study may contribute to the field of information technology by exploring how Master Data Management (MDM) and Service-Oriented Architecture (SOA) could enhance data quality and operational efficiency in the retail industry. MDM and SOA are crucial for enhancing data quality and business outcomes in the retail industry. MDM addresses poor data quality through tools, such as data quality rules, data survivorship instruments, data synchronization, data standardization, and interfaces for augmenting third-party data and maintain connections with standardization providers, while also building cross-references for merged data and creating golden records for integrated data (McKnight, 2006). The study recommended best practices for data management and integration, informed by the identified gaps, and factors contributing to previous implementation issues.

Key stakeholders, including business analysts and technical managers, played a critical role in addressing gaps in data management and leveraging the structured delivery model proposed in this study. By identifying gaps, IT practitioners specialized in MDM

and Integration Architecture can become more informed about the areas of concern that directly correlate with the implementation of these concepts. Insights from this research can help business analysts and technical design robust MDM frameworks and integration architectures to address past failures and standardize data solutions. Practitioners will gain the tools and insights needed to implement these concepts successfully in the retail industry's data domains by offering a detailed list of factors for project failures and introducing a standard model or delivery framework. This knowledge can influence the development of structured models and connect the identified factors to implementing MDM and SOA. The findings of this research will enable stakeholders to design a framework for integrating MDM and SOA to obtain customer insights that can enhance the retail industry.

Nature of the Study

The qualitative research methodology and basic action research (AR) were selected to address challenges in implementing MDM and SOA in the retail industry's customer domain. Cilesiz (2010) emphasized that studying individuals' experiences with technology and implementing IT frameworks provides insights into designing and improving future implementations. By leveraging qualitative methods, this study explored first-hand experiences to identify factors contributing to implementation failures. The AR design facilitated a systematic approach to problem-solving, including data collection, analysis, and actionable planning, ensuring practical solutions to the identified issues. This research design aligned with the study's objectives to explore and address the underlying causes of MDM and SOA implementation challenges in the retail sector.

Methodology

Qualitative research was effective in investigating MDM and SOA implementation failures in the retail sector by leveraging insights from experienced business architects and technicians. According to Nigar (2019), qualitative methods enable IT professionals to design effective data systems and support MDM implementation to achieve business objectives. The qualitative approach allowed for in-depth exploration of practitioners' experiences, revealing the underlying factors contributing to implementation challenges. By engaging directly with IT professionals, the study provided rich, detailed insights into data systems and their role in enhancing retail business structures. This method not only identified existing issues but also informed strategies to improve system implementation and achieve business objectives. Through qualitative research, this study advanced understanding of the practical and technical aspects of MDM and SOA implementation, offering actionable insights to improve outcomes in the retail sector.

Basic action research (AR) is a qualitative research design aimed at solving problems and improving professional practices. According to Saunders and Lewis (2019), AR focuses on addressing specific issues in professional settings. Reason and Bradbury (2013), described action research as a collaborative method involving researchers and practitioners, utilizing approaches such as interviews and discussions to foster participant engagement. Burnes (2009) emphasized that AR generates practical and theoretical insights while promoting continuous growth, self-development, and change. By fostering collaboration and focusing on continuous improvement, action research serves as an

effective qualitative framework for addressing complex problems and advancing professional practices.

This study leveraged only interviews on master data management and service-oriented architecture implementations. The researcher interviewed twenty business and technical managers with at least five years of experience in the implementation of MDM and SOA in the retail sector. Interviews were selected because the researcher needed to collaborate and understand the participants' experiences in SOA and MDM processes in the retail industry and create an interactive session to get relevant data based on the interview questions aligned with the research questions. Business architects and technical experts in management were asked how they apply MDM and SOA in data management and the challenges they encounter in implementing MDM and SOA architectures. The technical experts and business architects were asked about the changes in data management efficiency and the quality of data acquired before and after implementing SOA and MDM to evaluate the role of the architecture in improving data management.

Design

Basic action research leverages practical problem-solving and knowledge generation. Introduced by Kurt Lewin in the 1940's, emphasizing a participatory and iterative process (Reason & Bradbury, 2013). Checkland and Howell (1998) expanded the field by developing Soft Systems Methodology (SSM), a structured framework of ideas and methodology for planning and addressing complex issues. Goede and Taylorlien (2019) argue that SSM effectively supports action research by guiding its iterative cycles. This study adopts basic action research, leveraging interviews with a

sample group of information technology practitioners to address practical challenges while contributing to scientific knowledge.

Basic Action research (AR) is an evolving process to integrate applied behavioral science with organizational insights to address strategies, practical skills, and structured hierarchical systems, applying across different fields and various business functions and disciplines. It aims to drive organizational change and advance scientific knowledge through collaborative and shared exploration (Shani & Coghlan, 2021). Action research was appropriate for studying the risks and factors from past failures of implementing MDM and SOA in the retail industry because it combines a systematic research approach with practical problem-solving, collaborative inquiry, and a focus on context-specific understanding and application. These characteristics make it well-suited for addressing complex, real-world challenges like technology implementation related to customer domain for the retail industry.

The answers and data collected from the interviews helped to collaborate and identify the risk factors from past failures of implementing MDM and SOA within the customer domain for the retail industry as it relates to iterative processes. Interviewing provided an opportunity to discuss and link the information from the findings with previous research and literature. Using scholarly articles and white papers informed how previous implementations have been unsuccessful in the retail industry. Due to the research gaps, there was less available in correlation with data domains when searching and gathering literature.

Research Questions

Three questions were created for this study. Answers to the three questions provided relevant data to meet the study objectives. Notably, the following questions were used to review the implementation of MDM and SOA in the retail industry:

R1. What is the role of effective data management systems and a sound event-driven architecture in contributing to measuring and acquiring business insights?

R2. What is the impact of MDM and SOA on creating process improvement in quality and synergy of data across enterprises within the retail industry for the customer domain?

R3. What are the impacts of implementing effective enterprise-wide data solutions have in the retail industry?

Conceptual Framework

This section outlines the organization and purpose of the conceptual framework guiding this research. The framework establishes the connection between theoretical concepts and empirical inquiry, offering a structured approach to the study's design, data collection, and analysis. Unlike a theoretical framework, which primarily explains phenomena using established theories, a conceptual framework focuses on integrating and applying key concepts to the research context. As Ravitch and Riggan (2016) explained, conceptual frameworks identify key variables, relationships, and assumptions, ensuring alignment with research objectives and grounding the study in existing literature. This approach is particularly appropriate for examining the implementation of Service-Oriented Architecture (SOA) and Master Data Management (MDM) in the retail industry, where practical applications of theoretical ideas are necessary to address

industry-specific challenges. By using a conceptual framework, this research established a structural and practiced approach to exploring the relationships between data management practices and business outcomes. This focus ensured that the study remained coherent and relevant to both theoretical discussions and real-world applications. The conceptual framework provided the foundation for addressing the research problem, ensuring the study contributes meaningfully to the literature on SOA and MDM in retail.

The conceptual framework integrated theories and concepts to guide the analysis of SOA and MDM in enhancing retail industry outcomes. Conceptual frameworks facilitates data analysis by identifying relationships between variables and grounding research in theoretical insights (Ravitch & Riggan, 2016). For instance, (De Camargo Fiorini et al., 2018) emphasized that theories applied to disciplines such as information systems can support the study of architecture, data management, and governance in retail. However, research addressing the implications of these frameworks remains limited. This study addressed the gap by using research questions to examine how SOA and MDM influence productivity, customer service, and business outcomes. The conceptual framework drew on theoretical insights to structure the analysis, ensuring that findings are robust and relevant. This approach demonstrated the importance of integrating theoretical concepts into conceptual framework to enhance the study's coherence and contribution to the field.

Inductive theories within the conceptual framework demonstrated the practical applications of SOA and MDM in retail businesses. Inductive theories focus on addressing practical challenges such as data inconsistencies, inefficiencies, and governance issues within business infrastructures (De Camargo Fiorini et al., 2018).

These theories help bridge the gap between abstract concepts and their real-world applications in the retail sector. By applying these theories, the study explores how SOA and MDM systems enable businesses to better assess, implement, and analyze data management practices. This ensures that the research remains grounded in practical implications, addressing critical business needs. The inclusion of inductive theories enhances the framework's ability to provide actionable insights, contributing to both academic literature and industry practices.

The conceptual framework ensured the study's coherence and credibility while addressing gaps in the literature on SOA and MDM. Previous studies have shown that robust systems and infrastructures can optimize processes and improve outcomes in the retail sector (De Camargo Fiorini et al., 2018). However, the lack of research on the implications of data management in retail presents a significant gap. By grounding the study on a conceptual framework, this research ensures that the findings are not only theoretically informed but also practically relevant. This integration of theoretical perspectives strengthened the framework's evidence base, enabling meaningful contributions to literature. This conceptual framework provided a structured approach for exploring SOA and MDM, contributing to advancements in data management and governance in retail businesses. Diffusion of Innovation, Resource Based, and Dynamic Capabilities were referenced throughout this paper and were used to support the research study and business practices.

Diffusion of Innovation Theory

Diffusion of Innovation Theory identifies the process of adoption across five stages: knowledge, persuasion, decision, implementation, and confirmation. Baskerville

and Pries-Heje (2001) assert that the diffusion of innovation theory provides insight into how information technologies are adopted and how technological innovations spread within and between communities. This theory highlights the role of communication channels, social systems, and innovation characteristics such as relative advantage, compatibility, complexity, trialability, observability on adoption rates. The theory views innovation as a new or unfamiliar idea, technology, or process introduced within a specific social system or area.

Over the decades, the theory evolved to incorporate factors influencing organizational innovation, including leadership and environmental dynamics. Greenhalgh et al. (2004) explored its application in complex organizational systems. In recent studies, the theory has been applied to the adoption of big data analytics in retail (Dekimpe, 2020). Rogers' initial work focused on agricultural innovations, later expanding to various domains like healthcare, education, and technology. Another example includes the implementation of mobile payment systems in developing economies, where trialability and compatibility influenced consumer acceptance (Mbinkeu, 2014). The theory provides a relative advantage of data-driven decision-making encourages faster adoption among innovators and early adopters.

Diffusion of Innovation Theory is perceived as sharing information from one person to another within a social system. Santacreu (2014) stated the main determinants of successful innovation diffusion include the innovation's attributes, communication channels, the social system, and the attributes of the adopters. Communication channels include the medium used to acquire information about an innovation and usefulness. It considers interpersonal and mass media communication (Satchell, 2018). Innovation

attributes include complexity, observability, relative advantage, trialability, and compatibility. A relative advantage is the extent to which a user perceives improvements or benefits of using a specific technology. Compatibility focuses on a technology's consistency with the existing social and technical environments.

Complexity reviews the extent to which an innovation is hard to understand, use, or implement. According to Zhang & Zhang (2020), innovations can coexist or integrate with experiences and existing values and the needs of those who use them increase the prospects for adoption and diffusion. Pierpaolo (2011) stated innovation is considered less complex if users accept it easily. Trialability focuses on the innovation's ability to be tried with minimal investment and commitment. Innovation is perceived to have high trialability if it is more likely to be used by many end users. Observability is considered the extent to which an innovation's benefits are visible to the intended audience. According to Yu & Tao (2009), innovation is only adopted when it is beneficial to the end user. Therefore, an end user's perceived benefit from an innovation dictates the diffusion rate.

The diffusion of innovation theory characterizes the elements of a social system into five groups based on the attitudes toward innovation. The categories include early adopters, innovators, earlier majority, laggards, and later majority (Yu & Tao, 2009). Innovators represent a small part of the social system and are usually the first to implement new technology. Innovators can apply and understand complex technical knowledge which is vital to bringing innovation from the external part of the social system (Risman, 2018). Early adopters are more integrated into the social system, well-informed about innovation, more economically successful, and well-connected with the

new technology. Early and later adopters adopt technology after understanding the benefits. Laggards strongly resist new technology and can be perceived as non-adopters due to their lack of awareness about technology.

Resource Based Theory

The Resource-Based Theory (RBT) suggests that organizations can gain a sustainable competitive advantage by leveraging valuable, rare, inimitable, and non-substitutable (VRIN) resources. Introduced by Jay Barney in 1991, RBT emphasized the role of physical, human, or organizational assets that contribute to superior performance (Varadarajan, 2023). It originated as a response to the limitations of industrial organization economics, emphasizing the internal resources of firms rather than external industry structures. Foundational groundwork by Penrose (1959) and Barney (1991) established the theory, while later contributions by Peteraf (1993) and Grant (1996) expanded its relevance to dynamic industries, emphasizing resource heterogeneity and immobility (Helfat & Peteraf, 2009). This theory underscored the strategic importance of internal resources in securing long-term competitive advantage.

The Resource-Based Theory (RBT) emphasized that rare, valuable, and difficult-to-imitate resources drive long-term organizational success. According to Dutta (2013), strategic resources are the foundation for developing organizational capabilities that enhance performance. Miller (2019) expanded this by categorizing resources into business processes, assets, firm attributes, capabilities, information, knowledge, and distinguishing between internal and external resources. Internal resources include logistics, research and development capabilities, low-cost processes, and brand management, while external resources comprise customer demand, supplies, and

technological changes. In the retail sector, RBT explains how companies like Amazon gain competitive advantages through proprietary logistics and data-driven supply chain management (Miller, 2019). Similarly, RBT supports research on how Master Data Management (MDM) systems help firms outperform competitors by leveraging technological capabilities (Amajuoyi et al., 2024). RBT provides a framework for understanding how strategic resources drive competitive advantage, particularly in industries reliant on technological innovation.

The Resource-based Theory (RBT) explained how organizations leverage unique resources to achieve long-term competitive advantage and outperform others. Mulders (2019) noted that RBT assumes organizational resources are unique, serving as a key differentiator. Miller (2019), further explains that resources are heterogeneous, meaning organizations with distinct resources in particular contexts can perform activities more effectively and create a competitive advantage. This heterogeneity of resources enables firms to tailor strategies based on their unique assets, enhancing their ability to gain a competitive edge. By understanding and utilizing these unique resources, organizations can optimize their performance and achieve sustained success.

Dynamic Capabilities Theory

Dynamic Capabilities Theory, introduced by Teece, Pisano, and Shuen builds on RBT by emphasizing an organization's ability to integrate, build, and reconfigure internal and external competencies to adapt to rapidly changing environments. It highlighted three main dimensions: sensing opportunities and transforming processes. According to Denrell and Powell (2016), the dynamic capabilities theory was developed to improve the resource-based theory. The theory evolved as a response to the limitations of static

resource-based views in explaining organizational adaptation. More recent work by Helfat & Peteraf (2009) connected the theory to organizational learning and strategic management. The theory extends the arguments made under the resource-based theory organizations derive competitive advantage from unique capabilities. The theory posits that dynamic capabilities enable organizations to reconfigure, organize, and integrate their capabilities and resources to adapt to the changes in the business environment. Kodama (2018) stated organizations need technological flexibility and organizational flexibility capabilities to create a long-term competitive advantage.

Technological flexibility capabilities included selecting the technical knowledge associated with existing knowledge areas in the implementation of technology.

Organizational flexibility capabilities included the attributes of an organization's structure focused on the decision-making authority, information flow, and configuration duty. Jiao et al. (2010) stated organizations with high organizational flexibility can respond effectively to changes in their environment, thus creating a competitive advantage. Other dynamic capabilities that can be leveraged to create a competitive advantage include change and environmental sensing capabilities. Change capabilities included the ability to integrate resources into the organization to create and sustain change. Environmental sensing capabilities include understanding the business environment and how it can impact business operations. Dynamic capabilities can integrate, configure and develop resources while updating and innovating operational processes (Kodama, 2018). Therefore, dynamic capabilities are vital in establishing a strong competitive advantage and improving organizational performance and sustainability. MDM and SOA are dynamic capabilities because they can help the retail

organizations effectively manage customer data to generate insights based on customer purchasing behavior and reviews.

The study highlighted how leveraging SOA and MDM as unique capabilities can significantly improve the performance of retail companies and help gain valuable insights into customer behavior and needs, which can inform strategic decision-making. Liu et al., 2018 argued that implementing MDM and SOA enables retailers to gain insights into customer behavior, which informs strategic decision-making and fosters a culture of continuous improvement. This proactive adoption of MDM and SOA addresses current challenges while positioning retailers for long-term success in a competitive market. Dynamic Capabilities Theory further explained how firms like Walmart reconfigure their operations to integrate online and offline channels, addressing shifting consumer preferences (Peris-Ortiz et al., 2018). By embracing dynamic capabilities, retailers can align IT infrastructure with market demands, empowering them to thrive amidst uncertainty and change (Poelmans et al., 2008).

Definition of Terms

The research study used various technical terms readers may not be familiar with. This section defines the technical terms, which include Control Objectives for Information and Related Technologies, a data excellence framework, data governance, data quality, master data management, and service-oriented architecture.

Control Objectives for Information and Related Technologies (COBIT): Control Objectives for Information and Related Technologies (COBIT) is an IT governance framework designed to align strategy, define objectives, and establish policies for enterprise information management (Arthofer & Girardi, 2017).

Data Excellence Framework: Mastered data analyzed and cleanse to implement and integrate into the business infrastructure (Akoum & Bu Hazzaa, 2019).

Data Governance: Data governance is an organizational framework aligns strategy, defines objectives, and establishes policies for enterprise information (Chakravorty, 2020).

Data Quality: Data quality is the suitability of a dataset and dataset properties for a particular purpose and often depends on the data consumer using them (Nikiforova, 2020).

Master-Data Management (MDM): Master data management is a business inclusive of planning, monitoring, and provisioning tasks. It is used to ensure master data quality and appropriateness (Arthofer & Girardi, 2017).

Service-Oriented Architecture (SOA): SOA is an optimized form of technology architecture provides support for services, service compositions, and service inventories (Boukadi et al., 2018).

Assumptions, Limitations, and Delimitations

The research was influenced by various assumptions, limitations, and delimitations. This section outlined the assumptions, and the approaches used to deal with the limitations. Assumptions are the accepted truths considered by a researcher based on the certainty of occurrence in the research. Limitations are restrictions that could impact the success of the research study and are beyond the researcher's control. Delimitations are distinct from limitations and are referred to the specific boundaries or constraints the researcher intentionally sets to define the study.

Assumptions

Assumptions in qualitative research can vary based on the research study and design. One of the assumptions of this study is ten participants will consent and be a part of this research study. The second assumption is each research participant will provide detailed responses to the research questions. The third assumption is the same size will be appropriate for the research methodology. The fourth assumption is the research survey questions will provide the opportunity for open-ended answers. The fifth assumption is Service-Oriented Architecture and Master Data Management are effective data management tools in retail industry.

Limitations

The primary limitation of this study was time constraint imposed by the research timeline. The research is bound by a specific timeframe established by the supervisor, which restricts the duration available for various research activities. This constraint required the researcher to carefully adjust schedules for different tasks to ensure sufficient time for necessary revisions while adhering to deadlines. Effectively managing time within these limitations was critical to ensuring the quality and completion of the study within the prescribed timeframe.

Delimitations

The company and participants selected for this research study were contacted based on their experiences with implementing information technology systems for retail businesses. Only participants with at least five years of proven experience in implementing MDM and SOA were selected for the study. Experience will be proven by contacting the participant's colleagues. Information technology team members were

recruited and given the opportunity to self-select to participate. Based on the research methodology and design, the selected sample size would be appropriate for an implementation team. The researcher spent time collecting data from individuals and analyzing data results to provide a comprehensive data report of research outcomes.

Chapter Summary

Chapter 1 included the foundation for the paper, highlighting the study's significance and the qualitative data collection approach through the experiences of technology practitioners. It emphasized why the study is important, particularly in understanding the experiences of technology practitioners who play a crucial role in developing successful business systems within the retail industry. By highlighting the importance, the chapter sets a context for the reader, explaining how the findings can contribute to existing knowledge and practical applications.

Additionally, chapter 1 detailed the researcher's approach to gathering qualitative data. It explains the specific methods used to investigate the practitioners' experiences, such as interviews. Qualitative research is designed to capture in-depth insights and rich, detailed data quantitative methods might not fully uncover. By emphasizing qualitative dimensions and investigating the intricate and multifaceted nature, Chapter 2 reviewed relevant literature to provide context on the theoretical framework, data concepts, and methodology.

Chapter 2

Literature Review

Chapter 1 introduced the research proposal, background of the study, problem statement, purpose, and significance of the study. Chapter 2 provides background information on the implementations of Master Data Management and Service Oriented Architecture in the retail industry and reviews the conceptual frameworks and technology methods provides the basis for the study. Chapter 2 provided a comprehensive and critical synthesis of many research sources that have been added to the field of study. Chapter 2 was organized and provides a comprehensive outline inclusive of historical and current content to inform the study and methodology.

Chapter 2 provided the title searches and documentation criterion, historical concepts, history of data concepts, the nature of the retail industry including barriers, leadership influence and business culture and impact, SOA implementation, a synthesis of historical concepts, current content, current use of data concepts, a synthesis of current sources, theories relevant to the study, current challenges and solutions, methodology literature, and research design literature.

Title Searches and Documentation

According to De Camargo Fiori et al. (2018), various steps should be taken to search for articles, peer-reviewed journals, and IT websites. The keywords selected for the literature review were Service Oriented Architecture (SOA), Master Data Management (MDM), Big Data and Data Management. A library search through multiple scholarly databases such as ABI Inform Complete (ProQuest), Academic Search Premier (EBSCO), Business Source Premier (EBSCO), Chronicle of Higher Education (Chronicle

of Higher Education), Education Research Complete (EBSCO), ERIC (EBSCO), ScienceDirect, and Google Scholar and websites to search relevant documentation, content and information technology keywords from various publication sources such as research journals, peer-reviewed articles, books, and information technology (IT) websites. A level of rigorous and thorough key words such as action research designs, information technology, theoretical frameworks, management, and organizational theories, COBIT, master data management, service-oriented architecture, and implementation was used in various search engines to capture research has been done over the years.

Table 1 and Table 2 provided a breakdown of the number of peer-reviewed, government, industry, edited media, and internet sources used to get information related to the identified keywords based on the keywords used in most parts of the study. Therefore, the sources do not represent the number of sources used in the whole study.

Table 1

Keyword Search Terms and Resulting Reference Sources

Search Term	Peer-Reviewed	Government Source	Industry Source	Edited Media	Internet Source	TOTAL
Service Oriented Architecture (SOA)	16	0	0	0	1	17
Master Data Management (MDM)	9	0	0	0	5	14
Big Data	5	0	0	0	1	6
Data Management	12	0	0	0	3	15
TOTAL	42	0	3	0	15	52

Table 2*Keyword Search Term by Publication Date*

Search Term	2022- 2017	2016- 2010	2009- 2000	1999-1950	1950- older	TOTAL
Service Oriented Architecture (SOA)	3	5	9	0	0	17
Master Data Management (MDM)	8	5	1	0	0	14
Big Data	6	0	0	0	0	6
Data Management	12	3	0	0	0	15
TOTAL	29	13	10	0	0	52

The next section provided the historical content of information technology, and the frameworks have shaped the world of businesses and can maximize the capabilities to address the increasing demands for quality service from various enterprises.

Historical Content

Over the past decade, technological advancements have significantly improved the operations of retail businesses. Andary and Sage (2010), note that technological improvements help define a company's mission, meet consumer demands, and enhance data management in production. Rafeq (2019) asserted that data is the currency of the 21st century, with information systems enabling the accessibility and implementation of digital information. Amali et al., (2020) emphasized that effective implementation is crucial for achieving business goals and expanding organizational strategies. The evolving management of technology is essential to optimize information delivery and ensure smooth deployment with minimal flaws. These advancements highlight the critical role of technology in driving retail business performance and meeting strategic objectives.

Effective data management is essential for businesses to leverage data in decision-making and gain a competitive edge. Stedman and Vaughan (2019), defined data management as the ingestion, storage, organization, and maintenance, which enables informed operational decision and strategic planning. However, without proper framework, data becomes irrelevant (Rafeq, 2019). Bakshi (2017) noted that businesses increasingly rely on technology to capture customer attention, while Rafeq (2019) highlights the importance of theoretical framework to connect technology and information systems. A well-executed data management strategy enables businesses to gain competitive advantages and improve business performance (Stedman and Vaughan, 2019). But, the implementation of big data, while transformative, requires careful management to mitigate risks and address adoption challenges (De Camargo Fiorini et al., 2018). The integration of big data into business operations has the potential to revolutionize business practices and management but requires a strong data management framework to realize its full potential.

Data Concepts

Service-Oriented Architecture (SOA) represents the next phase in business development models, evolving from object-oriented models in the 1980's to component-based development in the 1990's, and now service-oriented models. Barry & Dick (2013) explained that SOA's emergence marks a shift from conventional application development to service-oriented models, with composite applications driving enterprise services. Erl (2005) further noted that SOA enables a shift from object invocation to message-passage among services. Barry & Dick (2013) described SOA's role in preserving the benefits of component-based models while promoting interoperability and

adaptability, as messages are exchanged between services without concern for the underlying execution methods. Oliveira et al. (2009) asserted that SOA builds on component-based architecture and distributed systems, addressing the need for loosely integrated, standard-based, and protocol-independent distributed computing.

The application of SOA in the retail industry has become increasingly critical due to its ability to enhance supply chain adaptability and coordination. SOA facilitated agility and interoperability in retail supply chains, addressing the demands of modern business models and globalization. Lee et al. (2010) highlighted how SOA reconfigures supply chains to adapt to evolving business needs and improves the coordination of supply chain components. Vernadat (2007) emphasized that supply chain agility depends on interoperability achieved through SOA implementation. In e-commerce retail, SOA enables the seamless integration of suppliers, partners, and customers, optimizing the performance of electronic supply chains. This agility is vital in meeting the growing demands of the contemporary retail sector. By leveraging SOA, the retail industry can create flexible and responsive supply chains, ensuring sustained competitiveness in a dynamic market landscape.

SOA enhances retail operations by integrating diverse systems and reducing software complexity. Lee et al. (2010) stated that SOA enables retailers to adopt profitable business models, integrates Point of Sale, labor management, store portals, SD, and RFID systems, and minimizes the need for costly software customizations. It supports collaboration among trading partners, optimizes operations, and accelerates the realization of merger-related revenues and costs. By facilitating resource sharing and maximizing the reuse of business services, SOA promotes responsive and scalable

operations. It extends the life of legacy systems, enhances data capture and analysis, and supports forecasting, planning, and allocation activities essential for retail competitiveness. Through its robust and adaptable framework, SOA empowers retailers to streamline processes, optimize resources, and drive sustained growth in dynamic marketplace.

The definition of MDM has transitioned. In 2005, MDM was defined as developing and maintaining master data and values and the relationship between them (Loshin, 2009). The definition changed by focusing on the research context and the perspective of researchers reviewing the concept. In 2011, the definition changed to bringing master data together to enable the application of master data management services such as data quality, data stewardship and governance, overall management of data life cycle, hierarchy, and metadata (Cervo & Allen, 2011). Researchers agreed MDM is the management of shared master data to reduce redundancy and guarantee better quality through a standardized reference with technology, governance, and process. It focused on serving data as a reference offering truth to consumers through consolidation and integration of master data from various data sources in a central system.

Integrating Service-Oriented Architecture (SOA) and Master Data Management (MDM) is crucial for enhancing data management, but it involves addressing and resolving significant challenges to support seamless implementation. Rehan & Akyuz (2020) highlighted that implementing SOA and MDM enhances data quality, playing a critical role in effective data management. However, challenges such as synchronization issues, data inconsistencies, handling temporal data and uncertainty complicate

implementation. Additionally, cloud computing offered scalability while introducing security risks and potential loss of control over applications and data. Successful integration requires addressing stakeholder concerns, especially regarding the impact of MDM on business processes. Data science also plays a key role in managing unstructured and semi-structured data, requiring stable and flexible MDM systems to support operational effectiveness. As companies increasingly adopt enterprise-wide SOA, the integration with MDM becomes vital for ensuring robust and reliable data management, thereby supporting organizational goals in a complex data-driven landscape.

Service Oriented Architecture (SOA) is a new type of IT system with specific structures in terms of flexibility and power. According to Cherbakov et al. (2005), it is based on a segmental basis such the components in an IT system can be decomposed and composed based on the diverse needs of the business users. For instance, business clients can create various IT solutions in different contexts or situations. According to Yousef et al. (2014), the customer plays a vital role during the implementation and planning of SOA systems to ensure it is tailored to the future and present organizational needs. In such a framework, the main feature is not the set of technologies that that should be employed but the services they offer to benefit business users' activities. Hasan (2004) stated the SOA concept can be interpreted based on different perspectives. From the business perspective, it can be viewed as a group of services that enhance an organization's ability to conduct business with suppliers and customers. When considering a technology perspective, it is a new project viewpoint including separation of modularity, concerns, composition, and service re-uses and a new programming technique based on tools and standards mainly include Web services.

Service Component Architecture (SCA) is a key aspect of Service Oriented Architecture (SOA), enabling the construction of composite applications through reusable services to meet specific needs. SCA is based on the principle that business functions are executed as services, which are assembled to create solutions. It focuses on components in a business application, including the services provided, their connections, endpoints, communication methods, and applied policies. The SCA Assembly Model defines how components are configured to form composites, with each component tailored to specific business functions. These components can offer services that influence business operations and add value by connection to other services within the assembly, facilitating effective integration and functionality. Thus, SCA provides the building blocks for constructing SOA solutions, allowing businesses to assemble modular, service-driven applications that meet diverse functional needs within a business process.

SOA offers a new way to design and conceptualize IT application folders, from an information technology (IT) viewpoint. SOA and other similar applications create a new sense of the phenomenon of IT outsourcing. Gilley (2000) argue conventionally, outsourcing IT meant the complete externalization of technology undertakings to a third party to reduce cost and take advantage of supplier competencies. This approach has various disadvantages. For instance, users do not have control of technology applications, and providers emphasize rigid, broad, standardized solutions. In the implementation of SOA, customers only rely, to some extent, on the service provider and preserve their own sense of business technological applications. SOA providers also provide more flexibility business clients may exploit to deal with various business issues. Gilley and Rasheed (2000) also argued based on these components, it is evident SOA is compatible with the

co-production of services, particularly because of the main role business customers play in planning the system and gathering or using various modules. SOA also offers an exciting case of co-producing value due to the possible influence of the service system related to how the client firm implements it and how well it improves the set of capabilities and resources of the business user.

The rise of specialized software systems such as SCM, ERP, EDI, and CRM has increased the cost and complexity, prompting the development of new technologies to facilitate information sharing across enterprise systems. Subramani (2004) stated that these integrating technologies support information sharing and business process automation, enhancing agility and performance. A study conducted by Malhorta et al. (2005) focused on the correlation patterns in supply chain connections and the generation of new information through the absorptive capacity system. Their research findings offer vital insights into the instruments in inter-organizational frameworks, the role of supply chain conformations, and information exchange. Another study conducted by Subramani (2004) focused on the benefits of using information technology in supply chains and established using IT in the supply chain plays an intervening role in the benefits of the management frameworks in the supply chain performance. The integration of these technologies allows for standardized data exchange across different applications, improving the efficiency of business processes and facilitating decision-making. By leveraging IT, businesses can enhance their supply chain coordination and responsiveness. These research findings demonstrate that integrating IT and modern technologies within supply chains is essential for improving business agility,

performance, and inter-organizational collaboration in a complex, interconnected environment.

There is a wide range of literature describing technology implementation in the retail sector, including the use of web services and Service Oriented Architecture (SOA). A study conducted by Hiu (2013) offered a hypothetical framework for examining the role of IT in business approaches and how new technologies are contributing to organizational flexibility. The researchers encouraged further research on how organizations acquire agility and the technologies result in flexible business models and processes. According to Chatterjee et al. (2002), using web technologies in organizations results in very valuable business process advantages. It supports the role of investment rationale, top management aid, and the level of synchronization in the integration of web technologies. Therefore, evaluate explicit phenomena linked to using interactive architecture and web technologies such as SOA and their influence on the performance of an organization.

Service-Oriented Architecture (SOA) provides significant benefits by enabling seamless integration across organizational boundaries, enhancing operational and supply chain performance. Chindove et al. (2017) highlight that SOA facilitates the integration of processes and data across clients, partners and business units by exposing business service to external stakeholders. Similarly, Murtaza & Shah (2004) asserted that SOA improves supply chain performance by serving as an integrative framework capable of assembling disparate technologies, data formats, and systems. By offering a platform for smooth collaboration, SOA enhances the interoperability of organizational systems and the efficiency of shared services. This integration improves information flow across silos,

enabling organizations to deliver cohesive services to end-users while fostering partnerships with external businesses. The ability to standardize and align processes across entities is particularly beneficial for optimizing supply chain operations. Consequently, SOA serves as transformative framework, empowering organizations to achieve greater connectivity and improved performance, particularly in complex, multi-faceted business environments.

The Nature of the Retail Industry

In some of the most recent academic literature, there has been some discussion on the advancement of information technology in big data predictions, analytics, and the extreme changes in the way retailers use data to inform the shopping experience. According to Dekimpe (2020), the retail sector is the ideal setting in which businesses can benefit from big data analytics. This market can gain a great deal when data is leveraged appropriately to help change the game in the retail world. A report conducted by McKinsey (2011) showed how big data could help to increase operating margins by up to 60% and cause an annual productivity gain of up to 1%. However, there are concerns about how retail businesses may invest greatly and abuse the use of data, causing negative returns. Therefore, there is a need for more research to be conducted to show the value of retailers using SOA and MDM.

Based on previous literature, there is still a strong need for research to inform the retail industry. It is not enough to have access to data and how it supports value creation but also how it can inform value appropriation (for the retailer and/or the partners) for big data investments (Dekimpe, 2020). Verhoef et al. (2016) shared how retail businesses should invest in data capabilities through various strategies. These strategies include four

dimensions such as people, systems integrate multiple data sources, data processes, and organizational structure. However, within the retail industry, there is a slow movement in the willingness to invest and implement data systems.

Retailing serves as a significant source of big data, offering extensive opportunities for creating business value through effective data management and technological implementations. Dekimpe (2020) emphasized that retail inherently generates vast amounts of data, which can be harnessed to unlock dynamic opportunities. Specific technologies like Service-Oriented Architecture and Master Data Management enable retailers to generate actionable insights and align these insights with financial outcomes. McKinsey (2011) identified 16 big data levers, across five functional domains that retail companies can leverage, including marketing, merchandising, operations, supply chain, and new business models. For instance, marketing involves customers micro-segmentation and enhancing multichannel experiences, while merchandising focuses on assortment and pricing optimization. Similarly, operational transparency, supply chain logistics, and the development of innovative business models are enhanced through big data. These domains illustrate the breadth of opportunities available when retailers integrate robust data management systems and technologies. By leveraging big data levers, retailers can optimize operations, enhance customer experiences, and drive profitability. Moreover, technologies like SOA and MDM are foundational to achieving these goals by enabling seamless data integration and fostering more informed decision-making processes.

Throughout history, retail companies have used data to drive business decisions. Dekimpe (2020) stated retailing is a big business that plays a major part in the economy

across countries through international trading, commerce and investments, manufacturing, production, policies, and even the use of information technology. However, some may suggest former data systems may cause issues when trying to move forward with new architectural structures are necessary for data management opportunities (Dekimpe, 2020). At the macro level of retailing, there is a great amount of data retail businesses need to collect, such as SKU numbers, transactions, or customer information (Dekimpe, 2020). Therefore, it requires data-driven customization driven by SOA and MDM.

Barriers to Data Access

Data access and delivery methods pose significant challenges, particularly due to security and privacy restrictions in data platforms, which can hinder the effective use of integrated data. Kurnia (2024) highlighted that security and privacy limitations in data platforms often prevent seamless data integration within Service-Oriented Architecture (SOA), pushing organizations to rely on alternative methods such as spreadsheets or batch processing. These alternative approaches can diminish the effectiveness of Master Data Management (MDM) by hindering the comprehensive utilization of integrated data, ultimately reducing the potential benefits of a robust MDM strategy. To fully leverage MDM and its association benefits, organizations must address these data access and integration challenges, ensuring that security and privacy concerns do not obstruct the flow and utility of data.

Leadership Influence

Leadership significantly influences the success and effectiveness of organizational strategies, including the deployment of Service-Oriented Architecture (SOA) and Master

Data Management (MDM) systems. Reddy and Gupta (2018) suggested that an assertive leadership style promotes such as data quality and integration. In contrast, a cautious leadership approach may result in partial implementation, limiting system effectiveness. Devineni (2021) added that strong leadership not only ensures the integration of clean, accurate data but also cultivates a culture that leverages this data for better business outcomes. Effective leadership drives the successful implementation of MDM and SOA by ensuring proper system deployment and fostering an organizational culture that values data integrity and utilization. This enhances data reliability and reporting capabilities, ultimately improving business performance. A well-executed MDM strategy, backed by strong leadership, enhances data reliability and reporting capabilities, thereby driving better business performance.

Leadership influences the topic because how an organization is run helps to drive the insights and results needed to help the business thrive. Leadership style allows one to understand how aggressive and or how complacent the implementation can or will be, meaning leadership can target all components of a SOA based solution or just components of it along with the components of MDM whether it is a full implementation of MDM (i.e., data quality, data integration,) to help get the results are needed. The influence from the problem statement will also allow the culture and organization to understand additional insights (which results in more business because of more sound data to report against) would come from a more robust and sound architecture encompasses clean and sound data as well as integration components.

Business Culture and Impact

The impact of the problem statement on organizational culture and employees largely stems from implementing Service-Oriented Architecture (SOA) and Master Data Management (MDM) systems. The transition to these systems often necessitates changes to existing processes and standards, which can lead to significant shifts in organizational practices (Choi et al., 2018). This change frequently triggers resistance to altering legacy processes, resulting in potential delays and complications during implementation (DPan et al., 2016). Such resistance highlights the challenges organizations face when integrating new technologies and adapting their workflows to align with MDM and SOA principles (Li et al., 2019).

Service Oriented Architecture (SOA) Implementation

Companies are embracing the use of SOA in their supply chain. Managers in different organizations have complained about the difficulty in demonstrating the value of SOA, thus slowing down the implementation. IT managers must decide on implementing SOA to facilitate the move to new technology podiums and to facilitate well-organized communication in the supply chain, but there is limited information on the impact of SOA in the organization, especially on the influence of SOA on the performance of the supply chain. According to Chindove et al. (2017), SOA included various types of innovations, using innovation in organizations, the diffusion of innovation in social systems, and the use of innovation at individual and group levels.

The critical part between adopters is the gap between the early majority and early adopters because the enthusiasm of the majority is necessary but not enough to create a sustainable use and acceptance of new innovations. Gronli & Bygstad (2012) stated that

hindering and facilitating factors are important in the management of the complex processes involved in SOA implementation. According to Chatterjee et al., (2002), the spread of technology-enabled marketing development such as customer tracking and monitoring customer behavior with statistical analyses and data mining elements is a personalization condition. Marketing management emerges from micro-segment, no differentiation, and single customer review and nano-segment. A micro-segmentation framework can be used to monitor purchasing decisions by developing micro-segments based on buying benefits and motives. According to Jiang et al., (2010), the micro-segmentation process identifies why customers make a purchase and put less emphasis on the customer. Therefore, consumer micro-segments are determined by the evaluative behaviors towards purchasing a service or product.

The micro-segmentation model enhances marketing efforts by linking customer behaviors to their underlying motivations, providing deeper insights into customer needs and interactions (Jiang et al., 2010). Jiang et al. (2010) asserted that this approach adds value by offering a more precise understanding of customer actions, which can be leveraged to tailor marketing strategies. The concept is expanded through nano-segmentation, incorporating both purchase and non-purchase behaviors to create more granular customer profiles. Nano segmentation is developed through phases such as infer, gather, track and nano-segment integrating both customer and behavior data. Granebring and Révay (2007) highlighted the role of Service-Oriented Architecture (SOA) in advancing nano-segmentation, as it supports data mining and customer tracking through a variety of external and internal information sources. An example of this approach is the Coop Pioneer concept, which combines sales and customer data to analyze the

transaction process, buying patterns, and marketing activities, demonstrating the practical application of SOA and nano-segmentation in optimizing customer interactions.

The concept of information sharing in SOA implementation has been reviewed in both information systems and operations management literature. A study conducted by Clemons & Row (1993) in information systems established the transfer of information using information technology has the exceptional ability to concurrently reduce an organization's operation and decision-making costs and channel partners' transaction costs. According to Mukhopadhyay et al. (1995), EDI systems offer a significant business value. A study by Ning & Yao (2023) established sharing information with suppliers and electric integration results in performance improvement. In operations management research, Murtaza & Shah (2004) established when only data about an order is shared across the supply chain, it misleads the upstream groups in their production and inventory decisions, leading to lower performance in the supply chain.

Studies in information systems and operations management demonstrate that increased information sharing within the supply chain enhances its overall performance by fostering coordination among supply chain members. Devineni (2021) explained that technological advancements have facilitated closer coordination among supply chain partners by enabling the efficient sharing of key information. The author categorizes this information sharing into five distinct areas of information sharing: sales, inventory, order status, production schedule, and demand forecast. Similarly, Murtaza and Shah (2004) contended that expanding information sharing in distributed supply chain leads to Pareto improvements in the overall supply chain performance. Kimchangbong and KimKyHyong (2013), emphasized the importance of transparency in information sharing,

as it helps partners make better decisions and prevents data misrepresentation, thereby enhancing coordination and reducing inefficiencies. These studies collectively highlight that effective information sharing, and transparency are critical for optimizing supply chain performance, improving decision-making, and fostering greater efficiency.

Information sharing in organizations has become more complex due to technological advancements that allow for tailored reporting, frequency of data access, real-time access, software integration, and access levels. This complexity requires dedicated IT infrastructure and labor for maintenance. Byström and Järvelin (1995) stated that the complexity of the information-sharing process leads to delays in information sharing, which can negatively affect performance. For example, the cost of inventory management decreases with the decrease in delays in the information-sharing process.

Tax complexity, including its impact on decision-making and information sharing, influences how individuals and organizations manage and utilize information. Hu et al. (2007) categorized tax complexity based on determinability, noting that tasks with predetermined components are less complex, while tasks with uncertain outcomes or processes are more challenging. Vakkari (1999), stated that complexity affects how users interact in groups, limiting information assimilation, and leading to decision-making errors. Byström and Järvelin (1995) further highlighted that the complexity of information sharing, particularly in supply chains, can negatively affect performance due to the increased pressure on supervisors to manage and interpret shared information. As tasks become more complex, the ability to manage and assimilate information diminishes, increasing the likelihood of errors in decision-making. In supply chains, higher complexity in information sharing forces greater cognitive effort, potentially

reducing operational efficiency. Understanding and addressing task complexity, especially in the context of information sharing, is crucial for improving decision-making and performance, particularly in environments like supply chains where the complexity of information sharing is a key determinant of success.

Synthesize Historical Sources

The historical sources discussed in this chapter were used to compare the information that supports the study's purpose and helps expose the research gaps. Historical research indicates data management has transitioned from object-oriented models in the 1980s to component-based development models in the 1990s before becoming service-oriented models in the 2000s (Hardgrave & Johnson, 2003). The definition of MDA has transitioned from being a process of developing and maintaining master data and values. The relationship between them in 2005 to bringing master data together to enable the application of master data management services such as data quality, data stewardship and governance, overall management of data life cycle, hierarchy, and metadata in 2011 (Hardgrave & Johnson, 2003). The changes in definition have also revolutionized the viewpoint of researchers on MDA and SOA and how the two architectures are integrated into data management. Researchers slowly transitioned from defining and breaking down the components of SOA and MDA to their implementation in different industries, including the retail industry.

Current Content

Effective data management is essential for retail businesses to remain agile, identify market trends, and capitalize on new opportunities. Kumar et al. (2023) stated that well-managed data enables businesses to improve decision-making, comply with

regulations, and identify potential cost-saving opportunities. However, poor data management leads to inconsistent datasets, data silos, and quality issues, which can hinder application performance and produce inaccurate results. A comprehensive data management process, including data processing, storage, governance, and the integration of operational and analytical systems, is necessary for maintaining data integrity and facilitating business operations. Stedman and Vaughan (2019) emphasize that building a robust data architecture is crucial as it organizes business data and provides a blueprint for deploying the required infrastructure and technologies to support large-scale data applications. By establishing a solid data architecture and effective management practices, retail businesses can enhance their data-driven capabilities, leading to improved operational efficiency, better market insights, and stronger competitive positioning.

Competition and staying relevant in the industry require businesses to stay informed on ways in which data can inform and produce successful outcomes in the retail industry. Talend (2021) explains that retail brands often face business challenges and require help to stay competitive. This is where master data management remains important. Access to data can help streamline systems, inform daily operations, maintain a competitive edge in retail, and leverage this information in the business world (Talend, 2021a). An example is when businesses can operate on service-oriented architecture and integrate various data sources, including business transactions, billing, protection related to fraud detection, and deliver a seamless customer experience (Talend, 2021a). In addition, a retail business can implement master data management to link data across systems to aggregate, normalize, and interpret data in ways that allow them to achieve business objectives and goals (Talend, 2021b).

A recent case study by Aversa et al. (2021) showed research on identifying the knowledge of, needs, and development of big data management environments as well as exploring the opportunities and challenges associated with new data-based decision support systems within retail organizations. This study provided the preferred data environment encompassing data management adoption, development, and implementation in a retail business (Aversa et al., 2021). Based on the findings, it was shown even with the knowledge and need for big data and implementation of data management systems, it is important for retail businesses to use data to make decisions and create an environment is transparent on business data strategies and goals (Aversa, et al., 2021).

In some of the most recent academic literature, there has been some discussion on the advancement of information technology in big data predictions, analytics, and the extreme changes in the way retailers use data to inform the shopping experience. According to Dekimpe (2020), the retail sector is the ideal setting in which businesses can benefit from big data analytics. This market can gain a great deal when data is leveraged appropriately to help change the game in the retail world. A report conducted by McKinsey in 2011 showed how big data could help to increase operating margins by up to 60% and cause an annual productivity gain of up to 1% (Dekimpe, 2020). However, there are concerns about the way retail businesses may invest greatly and abuse the use of data causing negative returns. Therefore, there is a need for more research to be conducted to show the value of retailers using SOA and MDM.

Use of Data Concepts

Across several industries, common data concepts are fundamental and shareable across a business enterprise. Valuates Reports (2020) stated there is a rise and strong demand for comprehensive data infrastructures and data quality tools can improve operational excellence. Tools can be used for data quality can help to decrease data replication, minimize rework, and maintain reliable records of data (PR Newswire, 2020a). Retail businesses can leverage the high demand for Master Data Management solutions to show the value of data quality and data consolidation. MDM solutions combined with data quality tools for efficient data help improve customer support and optimally manage financial-based business risks (Valuates Reports, 2020).

Master data management (MDM) continues to evolve as the needs of enterprises continue to change and demand specific services can produce more effective outcomes. According to Vilminko-Heikkinen and Pekkola (2015), master data management provides value within a company and can improve how data can be used effectively through information technology systems. Master data management aims to solve data management issues by focusing on the processes of how data is injected and consumed from an enterprise, the quality of the data, as well as standardization, and use of integration (Vilminko-Heikkinen & Pekkola, 2015). Vilminko-Heikkinen and Pekkola (2015) explained the overall goal is to identify the key failures often come from data fragmentation, unreliable systems and processes, and complex infrastructures. If implemented appropriately, MDM can solve plenty of issues for an organization.

Service-Oriented Architecture (SOA) is a rapidly evolving IT concept that is increasingly being adopted by supply chain businesses and enterprise information

systems, though challenges remain in its implementation. According to Mohammadi and Mukhtar (2017), service-oriented architecture (SOA) has the potential for widespread adoption in supply chain business models and enterprise information systems. Since the early 2000s, there have been many discussions on the misguidance of how to implement SOA successfully, with unanswered questions about SOA's impact on organizational performance and business values (Joukhadar & Rabhi, 2016). Failures in SOA governance are commonly observed, and retail businesses often struggle with consistency and automation in IT and data management (Precisely Editor, 2020). Despite these challenges, SOA remains a powerful tool for retail enterprises, offering an effective architectural framework where systems function as service users and providers (Andary & Sage, 2010). To maximize SOA's impact, further research and improvements in governance and implementation strategies are necessary.

Synthesizing Current Sources

The integration of Master Data Management (MDM) and Service-Oriented Architecture (SOA) has the potential to improve enterprise systems, though challenges remain in their effective implementation. Berson and Dubov (2010) described how master data management and service-oriented architecture create synergies to deliver data from the master data management hubs to business application systems and users through services, shaping how information is shared and made accessible. Despite these benefits, challenges and failures arise in integrating these frameworks. While MDM and SOA offer promising solutions for improving data accessibility and enterprise functionality, their successful application requires overcoming governance and integration challenges. The Diffusion of Innovation, Resource-based and Dynamic Capabilities theories will

provide a foundation for understanding and addressing the challenges in the research study. This research aims to build on existing literature and frameworks to explore how users experience these technology systems and how their integration can be optimized to deliver desired outcomes in enterprise data management.

Theories Relevant to Study

The main theories informing this research study are diffusion of innovation, resource-based, and dynamic capabilities theories. These theories were referenced throughout this paper and provided a foundation for the study. The researcher will review research on applying theories implementing data management frameworks and how the elements may influence the implementation of SOA and MDM in managing data in the retail sector. The discussion below provides an overview of how each theory was applied in the development of the research questions.

The Diffusion of Innovation Theory

The diffusion of innovation theory played a pivotal role in facilitating the adoption of MDM and SOA among retailers. This theory aligned with RQ 2 and addresses the challenges and process improvements associated with implementing MDM and SOA. It offered valuable insights into how these innovations can be effectively introduced and embraced at various levels within the retail industry (Mbinkeu, 2014). By highlighting key factors that influence the adoption of these technologies such as relative advantage, compatibility, and complexity, this study underscored the importance of understanding these elements when developing strategies to mitigate resistance to new technologies. Addressing the challenges associated with the adoption process, organizations can reshape their policies and practices to create a more conducive

environment for innovation (Saghafian et al., 2021). For instance, emphasizing the relative advantages of MDM and SOA can encourage stakeholders to recognize the benefits, such as improved data quality and enhanced system interoperability. Additionally, ensuring compatibility with existing systems and processes can alleviate concerns about complexity, moving smoother (Saghafian et al., 2021). Leveraging the insights from the diffusion of innovation theory can also help retail leaders implement change fostering a culture that embraces new technologies. This proactive approach minimized resistance and enhanced organizational agility, enabling retailers to respond swiftly to market demands and improve overall operational efficiency (Wisdom et al., 2013). By changing processes and influencing organizational practices, retailers can position themselves for sustained growth and competitive advantage in an ever-evolving landscape (Wisdom et al., 2013). Therefore, the diffusion of innovation theory is an important part of integrating innovation such as SOA and MDM in the retail industry.

The Resource-Based Theory

The Resource-Based Theory aligned with RQ 1 and will be employed to illustrate how MDM and SOA can serve as strategic resources that provide a competitive advantage in the retail sector. This theory emphasized the importance of leveraging valuable and inimitable resources, such as well-integrated data management systems, to enhance operational efficiency, reduce costs, and improve decision-making processes (Creswell & Clark, 2023). Changing organizational processes to prioritize the integration of MDM and SOA will help retailers address existing inefficiencies and data silos that hinder performance. This shift streamlines operations and fosters a culture of data-driven decision-making, enabling leaders to make informed choices that align with business

objectives (Creswell & Clark, 2023). The research demonstrated how effectively utilizing these systems allows organizations to craft robust business strategies that capitalize on their unique resources. Retailers can create a more agile and responsive operational framework by influencing organizational policies and practices to embrace MDM and SOA. Amajuoyi et al. (2024) argued that the agile approach enhances long-term organizational success and positions the company to adapt to market changes and customer demands more effectively, ultimately driving sustained competitive advantage. Therefore, SOA and MDM were considered important resources in implementing the agile approach in the retail industry.

The Dynamic Capabilities Theory

The Dynamic Capabilities Theory aligned with RQ 3 and focused on how retail organizations can adapt and reconfigure their resources in response to changing market demands. This theory underpinned the study by illustrating how innovations can be integrated into retail operations to enhance an organization's ability to sense and respond to evolving market drivers (Peris-Ortiz et al., 2018). Adaptability is crucial for maintaining a competitive advantage, as it enables organizations to respond proactively to market changes rather than reactively (Poelmans et al., 2023). By changing processes to incorporate MDM and SOA, retailers can foster greater flexibility, allowing them to quickly adapt to new technologies and shifting consumer preferences in a dynamic environment.

Conceptual Framework Literature

Data management is a master challenge in the retail industry due to the data gathered from customers, suppliers, and vendors. Whether a framework is inductive or

deductive depends on the approach to reasoning and development. According to Hyde (2000), quantitative research focuses on deductive reasoning based on general principles, theories, or hypothesis testing, which can be applied to specific situations or factors. However, inductive frameworks are based on specific observations, patterns, or data points seeking to establish generalizations, new theories, or hypotheses (Hyde, 2000).

The Control Objectives for Information and Related Technologies (COBIT 5) is a deductive business framework used to interpret the data management process used in the retail industry. The framework is classified as a conceptual framework in this research to serve as a roadmap for developing the arguments about the improvement, monitoring, and development of data management and IT governance. COBIT 5 provides a structured approach for data governance and management, detailing processes, controls, and objectives. The primary aim is to develop or test theoretical models but offers practical guidance and standards for implementation.

A conceptual framework organizes and presents concepts and practices to guide data governance and data management rather than testing or developing theoretical explanations. COBIT 5 was developed by ISACA and designed to be tailored based on the needs of any enterprise (Kulkarni & Che, 2019). The COBIT 5 framework simplified several managerial procedures and combined processes, including inputs and outputs, project objectives, key process tasks, elementary and maturity models, and performance measures (Invensis Global Learning Services, 2018). COBIT 5 gave enterprises a significant advantage in implementing systems using concepts that would increase productivity and improve service quality. Tugas (2010) explained COBIT was once an

auditing tool to assess IT governance but has evolved into an information management resource.

In November 2018, COBIT 2019 was introduced to the world to leverage new advantages and programmatic upgrades (Kulkarni,2019). Rafeq (2019) explained how COBIT framework is not designed as a one-size-fits-all governance system for an enterprise. Retail enterprises are unique and, based on the size, may require specific stakeholder needs, mission, vision and outcomes, business and organization processes, technology, compliance requirements, and management styles to be addressed (Rafeq, 2019). Therefore, during the implementation phase, several key factors should be considered. Rafeq (2020) identified key factors: distinct enterprise character and profile, size, industry sector, regulatory landscape, threat landscape, the role of information technology in the company, tactical technology-related choices, and other relevant attributes (p.4).

COBIT is deductive and offers an integrated governance and management perspective empowers individuals to be innovative when building new data platforms, information systems, and business models (Rafeq, 2019). Amali et al. (2020) identified information governance as the responsibility of leaders and consisting of organizational structures and processes that can ensure the sustainability of the enterprise. Amali et al. (2020) added the effects of using best practices such as planning and organizational tools, delivery, and support, as well as supervision and evaluating technology to ensure efforts align with business goals. Rafeq (2019) acknowledged how current enterprises also require employees to contribute by remaining knowledgeable of the design functions of

the framework and building the capability to apply systematic approaches and methods effectively.

COBIT 2019 has evolved to meet the diverse needs of multiple users within and across enterprises. COBIT 2019 is useful in businesses such as retail with multiple departments and locations (Rafeq, 2019). However, users often report the system is complex and too generic, leading to a great deal of misunderstanding. Often enterprises will use COBIT to meet specific needs and not acknowledge to maximize full capacity. Implementation requires an assessment of enterprise needs to customize the COBIT content and determine relevant best practices to adopt (Rafeq, 2019).

A design guide for information technology governance solutions was developed to address some concerns. The design guide is customized to support dynamic governance systems and tailored to specific enterprise needs and supports (Rafeq, 2019). Amali et al. (2020) explained most maturity models were developed for information technology governance are a response to the existing need for processes and structures. However, COBIT is the only framework that emphasizes the efforts on IT governance. Rafeq (2019) believes the COBIT governance system is so dynamic and designed to consider other important factors such as technology or strategies that are impacted when changes are needed. The tailored enterprise needs are customized, and parameters are set to stay aligned to use the framework and prioritize data concepts (Rafeq, 2019).

Methodology Literature

Different methodologies test theories and investigate the effectiveness of informative systems and processes. Several studies utilized the benefits of qualitative methods. A qualitative study conducted by Hovmøller and Tambo (2014) found MDM

connects data related to individual business processes along the product life cycle and value chain, thus enhancing data management efficiency. According to Fleckenstein & Fellows (2018) the best MDM architecture should include seamless integration in an environment with various coexisting systems, an overview of all data connected to one product and the possibility to looking for a product based on product attributes.

Qualitative research methodologies are effective in exploring the impact of Master Data Management (MDM) and Service-Oriented Architecture (SOA) on data management and customer insights across industries. Cilesiz (2010) highlighted the widespread use of qualitative research to study life experiences. According to Teherani et al. (2015), stated that a qualitative research study is the best methodology with limited prior research on the topic. Zhang and Zhang (2020) found that MDM enhances data accuracy and decision-making outcomes within the health sector. Similarly, Smith and McKeen (2017) demonstrated that integrating MDM and SOA improves data accessibility and customer relationship management in the financial sector. Nguyen et al. (2018) illustrated in a case study how a leading company in retail sector operates using MDM and SOA to gain insights on customer behavior and improve marketing approaches. The case study found that MDM and SOA are effective in gaining customer behavior insights leading to better decision-making. Existing literature underlines the relationship between MDM and SOA and data governance; there are gaps in the literature on the use of SOA and MDM in gaining customer insights across industries. The current research on the research topic has used case study research methodology to demonstrate the effectiveness of MDM and SOA in data management in the retail industry. Although

the research demonstrates the integration of MDM and SOA in data management, there is a need to expound on the use of SOA and MDM in gaining customer insights.

Research Design Literature

Research design provides a structured approach to integrating various components of a study, guiding the collection and analysis of data. Saunders and Lewis (2019), defined research design as the strategy used to organize and connect the research elements chosen in a logical and coherent manner. Cook and Cook (2016) identified several types of research designs, including action research design, cohort, causal, descriptive, cross-sectional design, exploratory and experimental design, longitudinal, historical, philosophical, observational, case study, and sequential design. Action research, specifically, is a qualitative methodology where practitioners conduct research to address real-world problems in specific situations (Saban, 2021). The choice of research design significantly impacts how data is collected, analyzed, and interpreted. Action research, for instance, is particularly effective for solving practical problems within a specific context, making it suitable for studies aiming to address organizational or situational challenges. The study will leverage an appropriate research design, such as action research, to explore and resolve issues within the context of MDM and SOA integration, thereby providing actionable insights for practitioners in the field.

Action research was well-suited for exploring complex issues like MDM and SOA implementation because it emphasizes collaboration, practical problem-solving, and a deep understanding of participants' perspectives. Action research design provided researchers who are actively involved and affected by a problem with the ability to find a solution and provided feedback on outcomes (Saban, 2021). Action research emphasizes

collaborative inquiry and transformative change, is rooted in critical reflection, and combines practical problem-solving with theoretical research. Therefore, action research is appropriate for research questions or problems focused on providing solutions to improve practice (Zuber-Skerritt, 2016). For instance, in the current research, information on the challenges encountered in the implementation of MDM and SOA in the retail sector and effective implementation practices to improve the practice of using MDM and SOA was gathered from business and IT experts to gain business insights in the retail industry.

The case study design includes conducting a detailed study of a specific research problem. According to Crowe et al., 2011, this design can help to narrow down a broad research field into a few examples easily researched. For example, the researcher may review case studies on implementing MDM and SOA in the retail industry to identify effective practices, challenges, and improvement areas. The case study design is appropriate when the researcher needs to obtain a detailed appreciation of an event, issue, or interesting phenomenon in its natural context.

A cohort design includes conducting a study for a period by involving research participants from a population from which a representative member or subject comes and who are connected by some similarity or commonality (Cook & Cook, 2016). According to Siegel and Stenson (2014), cohort studies can either be open or closed. An open cohort study involves a population defined by the state of being part of this research and is monitored for the research outcome. Cohort studies are appropriate when the outcome of the research can be predicted using predictive outcomes (Cook & Cook, 2016). But

closed cohort studies involve participants who join research at a defining point where presumed no new study participant can join the cohort (Siegel & Stenson, 2014).

A descriptive research design is appropriate when the researcher intends to identify attributes, categories, trends, and frequencies. Descriptive research design helps answer questions of what, who, where, when, and how associated with a specific research problem (Singh, 2009). The cross-sectional design is characterized by reliance on existing differences, no time dimension, and groups selected based on existing differences. The design focuses on measuring differences between or among various subjects, people, or phenomena (Cook & Cook, 2016). The cross-sectional design is appropriate when the researcher is population-based surveys.

Experimental research design is a quantitative research design used to predict or determine what may happen. When using the experimental design, the independent variable is administered to the experimental group, and the control and experimental groups are measured on similar dependent variables (Cook & Cook, 2016). The experimental research design is appropriate when the researcher intends to examine a cause-effect relationship. Therefore, this research will not use the experimental research design because the study is qualitative. The historical design includes collecting, verifying, and synthesizing evidence from past studies to develop facts to refute or defend a hypothesis.

Historical research design is a qualitative research method that systematically investigates past events to understand their causes, impacts, and relevance to contemporary issues. Historical research primarily gathers, verifies, and synthesizes evidence from the past (Hester, 2018). This approach is crucial in various fields,

including history, sociology, anthropology, and political science, as it helps reconstruct and interpret historical narratives through careful analysis of primary and secondary sources. This process allows researchers to establish facts supporting or refuting a hypothesis about historical events. By examining past occurrences, researchers can gain insights into social norms, cultural constructs, and values, thereby enhancing the understanding of how historical events shape present and future contexts (Hester, 2018). Therefore, a historical design is appropriate when the researcher intends to synthesize data from many sources.

Exploratory research design is a qualitative approach used to investigate problems that are not well-defined or understood. According to Cash et al. (2016), exploratory design involves investigating research questions that have not been studied. This research is essential in various fields, including marketing, social sciences, and product development, as it helps researchers gain insights and identify variables that may not be immediately apparent (Cash et al., 2016). The primary goal of exploratory research is to gather preliminary information that can help define a problem and develop a hypothesis. By using exploratory research, organizations can better understand customer needs, market trends, and potential challenges before committing to more extensive research or product development. Therefore, the exploratory research design is appropriate when the researcher is studying a new issue or when there is a challenging data collection process.

Additionally, the philosophical design includes using augmentation tools derived from philosophical concepts, traditions, models, and theories. According to Levers (2013), this design critically challenges and explores the relevance of evidence and logic in academic debates to analyze arguments about important issues or to discuss the cause

of existing debates about a research problem. The philosophical design is appropriate when the researcher intends to reveal the assumptions. This can lead the choices applied to the methodology and purpose design, data interpretation, and analysis.

Sequential research design combines longitudinal and cross-sectional methods, integrating quantitative and qualitative data to provide a comprehensive understanding of a phenomenon. The researcher analyzes the sample to determine whether to accept the alternative hypothesis, null hypothesis, or select other researchers' hypotheses and repeat the study (Cook & Cook, 2016). Sequential research design includes explanatory and exploratory designs. Sequential explanatory design is a two-phase design where the initiating phase includes collecting and analyzing quantitative data, and the next phase includes using qualitative data to explain or build upon the initiated quantitative data. Sequential exploratory design starts from the qualitative research phase where data is collected, followed by a quantitative phase in which the initial findings must be tested or generalized. The approach is well suited if the new variables or constructs must be established before they can be quantitatively measured (Creswell & Clark, 2023). Sequential research design is helpful when the problem under investigation requires an in-depth exploration of the phenomenon with further generalization or aims at the explanation of unexpected results derived from the quantitative phase (Plano Clark & Ivankova, 2016). Such studies allow the iterative refinement of research questions and hypotheses which is why they have become powerful tools in complex research settings where both quantitative data and contextual understanding are needed (Kroll & Neri, 2022). Therefore, sequential research design can be used when the research requires using both quantitative and qualitative research methods.

The causal design includes understanding a phenomenon based on conditional statements to measure the impact of a specific change on assumptions and norms. The causal research design aims at finding out how one or more variables, the independent variable(s), influence another variable, the dependent variable, directly. In causal research, different experiments are conducted to manipulate the independent variable. The researcher then observes the downstream effect of the independent variable on the dependent variable to infer the cause. Causal research design has thus been significant and highly applicable in the social sciences, marketing, and health fields. For instance, an individual may use causal research to determine if one type of marketing results in one more sale. This design typically involves a highly controlled setting that rules out any confounding factors that can affect the results yielded; in which case an assurance is attained that the effect is caused by the manipulation of the independent variable (Shively, 2017). A key strength of causal research design is that it can produce valid evidence of causality for making informed decisions in policies, businesses, and scientific research. However, its implementation might be challenging because this design needs to ensure unrelated variables are tightly controlled and most often demands experimental or quasi-experimental designs.

Conclusions

Using theories to discuss concepts such as master data management (MDM) and service-oriented architecture (SOA) has major benefits. As the literature has been reviewed, discussed, and synthesized, one can acknowledge the detailed efforts made to address how historical and current content informs the research. The information brings significant value to the technology field and how companies, researchers, and

practitioners have taken the time to evaluate how systems are designed and evaluate the instances in which failures occur. With this known information, researchers have taken the time to not only address these challenges but have also provided effective solutions. According to Rafeq (2019), it is through these solutions effective IT governance systems, processes, implementation practices, management, and a maturity model can be used to reassure quality and minimize errors. Master data management and SOA are effective in data management in the retail industry because they address the current data management challenges in the retail sector. For instance, master data management aims to solve data management issues by focusing on how data is injected and consumed from an enterprise, the quality of the data, as well as standardization, and use of integration. SOA defines nano segments because data mining and customer tracking technologies used with various external and internal information sources are necessary to understand customer insights and apply them in improving products and services.

Chapter Summary

A grounded literature review provided a strong basis for any research study. Most of the information in this chapter, from various sources, allows for a deep analysis of how this information can support the findings of the study. Much information has been provided to show the significance of research and qualitative research methods. The overall chapter discussed the opportunity for new research to be conducted and investigates the day-to-day experiences of people who work in the technology field to implement effective systems. The methodology and research design literature show the significance of using the most appropriate methods to collect data. It is a key component of a research study to capture essential information and analyze the findings within the

study. Several research studies have used qualitative methods to ensure individuals can speak to and express their experiences with technology systems. Therefore, this qualitative methodology was chosen for this research study. The next chapter presents the methodology and research design.

The literature discussed throughout this chapter brings great value to the industry and the research study presented. Throughout the information, there is a consensus there is a strong need to deliver quality services even at the basic levels of an enterprise. These efforts are important; however, there is a requirement leaders and employees of these organizations stay informed of the advancement of information technology systems and learn to use them to their full capacity. When this does not occur, there is a potential for challenges and a need for more research to address the implementation and technological improvements.

Chapter 3

Research Methodology

This qualitative action research study understood how the implementation and use of master data management and service-oriented architecture may increase productivity in the retail business industry. Businesses owners work to ensure they can produce quality services and make informed business decisions using data and technology infrastructures. Chapter 2 reviewed historical and current sources informing the need for the research study, the methodology, and research design. Chapter 3 provides a clear overview, and detailed research study approaches. The action research approach will be used to further investigate and use the experiences of IT professionals and their approach to informing the effectiveness of data and building systems are useful in everyday practices. Additional information will be in Chapter 3 is the research method and design appropriateness, research questions and instrumentation, population and sample, consent, and confidentiality.

Research Method and Design Appropriateness

A qualitative research method was selected for this research study due to its suitability for addressing research questions, which aim to understand specific concepts and experiences related to the implementation of SOA and MDM in the retail sector. Qualitative research was appropriate as it emphasizes gathering first-hand information through interviews, which is essential for analyzing the experiences of individuals in the retail sector. Singh (2019) stated that qualitative action research enables the researcher to collect data that justifies and proves the effectiveness of recommended actions. This approach was particularly appropriate for this study, as it focused on exploring solutions

to a current problem in the retail industry through data collection, organization, analysis, and interpretation. Action research was chosen for its ability to facilitate a systematic process that starts with problem identification, followed by data collection, organizing, analyzing, interpreting the findings, and finally developing and evaluating a plan based on these results. By using this method, the study not only addressed the problem but also bridged the gap between professional practice and theoretical concepts. Additionally, triangulation and member-checking were employed to minimize researcher bias, ensuring the accuracy and objectivity of data interpretation and analysis. The qualitative action research methodology was integral in exploring the implementation of SOA and MDM, helping to address a significant issue in the retail industry while enhancing the study's validity through rigorous data validation techniques.

Research Questions

Three research questions guide the study. The questions were used to gather and analyze data linked to the research topic and ensure the research topic was addressed. The research questions guiding this study are:

R1. What is the role of effective data management systems and sound event-driven architecture in contributing to measuring and acquiring business insights?

R2. What is the impact of MDM and SOA on creating process improvement in the quality and synergy of data across enterprises within the retail industry for the customer domain?

R3. What are the impacts of implementing enterprise-wide data solution have in the retail industry?

The questions were based on how businesses can use master data management and service-oriented architecture systems to be better equipped to produce business outcomes increase productivity, business sales, and customer service experiences.

Population and Sample

There were 3000 people eligible to participate. However, 20 participants were purposively sampled based on their expertise in MDM and SOA implementation within the retail industry. The criteria for participation included business and technical experts who were 30 years or older and had five or more years of experience in data management within the retail industry, encompassing the implementation of Master Data Management (MDM) and Service-Oriented Architecture (SOA). Research participants were recruited from a company that publicizes information on Master Data Management (MDM) and Service-Oriented Architecture (SOA). Recruitment outreach included professional colleagues with similar backgrounds in master data management implementation, in some project capacities. The list of potential participants was publicly accessible via the intranet. Therefore, a data access and use permission form is not needed.

Informed Consent and Confidentiality

In this section, the principles of informed consent and confidentiality are outlined to ensure that all participants in the research study understand their rights and the measures taken to protect their privacy and identity. Informed consent is a voluntary agreement to participate in the study, and participants are provided with detailed information regarding the purpose, procedures, risks, and benefits of the research study. Confidentiality protects and ensures any personal or identifiable information is secured,

stored, and only accessible to the researcher. Ethical considerations foster trust and transparency with the participants.

Informed Consent

Informed consent is a basic component in observing ethics in research. According to Nijhawan et al. (2019), informed consent is a process through which a participant or patient is informed of the procedures, risks, benefits, and alternatives in a proposed research or treatment before they agree to participate voluntarily. Informed consent is important for maintaining respect for participants' rights, dignity, and autonomy. It also acts as a reaffirmation of the willingness to be involved in the study or treatment. The researcher received informed consent from the research participants by requesting them to sign an informed consent form. The participants signed consent forms via email, as a pre-condition for participation. The informed consent was signed electronically using DocuSign to ensure participants agreed to the terms and understood their role in the study, and their rights were protected at all costs. The informed consent form included various components mainly, an introduction to the study, the purpose of the research, types of research intervention, participant selection, voluntary participation, research procedures, research duration, risks, benefits, reimbursements, confidentiality, the procedure to be used in sharing research results, right to withdraw or refuse and who to contact during the research.

Study participants were informed to not disclose their participation in the study to minimize any risk. These individuals are also informed at any time they can opt out of the study and no longer participate. Minimal risk to participants were addressed by conducting the interviews online and maintaining the anonymity of the participants. This

means the researcher will protect and respect the participant before, during, and after the study is completed. The researcher made every possible effort to decode, store, and destroy data appropriately to mitigate the risk of compromising participants' confidentiality.

Confidentiality is the ethical and legal responsibility one owes to others, which obligates a person to standards binding them to ensure certain privacy-protected information shared by others during interaction is not accessed by unauthorized persons. Confidentiality fosters trust, which is important in research. Confidentiality (pertaining to data management) included protecting and securing personal data through encryption and removing any personal identifiers. The confidentiality of the research participants was maintained by calling them research participants or respondents, such as Participant 1, Participant 2, etc. rather than using their names. The data was segregated and had no identifiers because the participants' names and gender will not be included in reporting the research findings. Any Personally Identifiable Information (PII) and study data collected were stored separately on an external encrypted drive and locked with passwords. The confidentiality of the data gathered from the researchers was maintained by encrypting the data and storing it in a password-protected folder. Recordings were destroyed/deleted after transcripts are reviewed and approved by participants. Study materials with any personal identifiers will be destroyed/files deleted after three years.

Instrumentation

For this study, interviews were the qualitative tool used to gain feedback from professional practitioners on the effective use of technology and informational systems for retail businesses. The researcher was the primary researcher engaging with the

research participants directly through interview sessions and directly sending informed consent form to research participants. The role of the researcher was to design the interview questions and select a venue to conduct the interview. The interviews were conducted online via zoom. The researcher elaborated on questions that were not clear and initiated discussions to contribute to gathering information.

During interviews, open-ended questions allowed for answers and discussion on experiences of data usage and the failures occur when MDM and SOA are not utilized and implanted within the retail industry. Through personal experiences with the research participants and by understanding how previous failures have occurred during implementation, the information can be streamlined to provide a framework on how to master these concepts. This was done using an enterprise-wide data solution of MDM and SOA and effectively measuring data within the retail industry. A structured model was proposed to address the need of the study based on the information and results from the interviews. The model helped standardize all the requirements of an enterprise-wide data solution for the customer domain with the implementation of MDM and SOA within the retail industry.

Open-ended questions were used to capture the experience of IT professionals who can inform how data in a retail business can be used most effectively. Greener (2011) stated open-ended questions are appropriate when a researcher is looking to learn and gather information on a particular topic has the potential to a variety of answers and helps to avoid having the researcher's beliefs and ideas imposed upon the participant. An effective approach was to create specific sub-questions which will help explore one's

experiences and inquire about detailed information from each participant. Table 3 below shows the research questions and interview sub-question alignment.

Table 3

Research Questions and Interview Sub-Question Alignment

R1. What is the role of effective data management systems and a sound event-driven architecture in contributing to the process of measuring and acquiring business insights?	Questions 1 through 6 are open-ended interview sub-questions developed to discuss the specific state of the retail company as it pertains to data, data management, data governance, data implementation, data analysis, and data outcomes.
R2. What is the impact of MDM and SOA on creating process improvement in quality and synergy of data across enterprises within the retail industry for the customer domain?	Questions 6 through 12 are open-ended interview sub-questions developed to specifically address this question as it pertains to effective ways IT professionals define data infrastructures and business practices to ensure and improve quality service, productivity, and customer satisfaction.
R3. What are the impacts of implementing effective enterprise-wide data solutions have in the retail industry?	

Field Test

The researcher conducted a field study before submitting the IRB application to collect individual responses about the fit and understandability of the questions for the exploratory qualitative study. Four designated experts were recruited to participate in the field study. One of the experts was a retail industry consultant with 15 years of experience in data management strategies. The second expert was the Chief Data Officer of a major retail corporation. The third expert was a professor of information systems specializing in qualitative research methodologies. The fourth expert was the Director of Data Governance at a retail technology organization. After the question interview guide was prepared, the respondent gathered the experts once again to analyze the interview questions and get their recommendations. One of the recommendations was that the

language used to phrase various questions should be simplified to better understand the questions. The second recommendation was that more probing questions should be included in the question set so more in-depth information about the views of the interviewees can be gathered. The third recommendation was that the sequence of some questions should be changed to enhance the interview flow. These recommendations were considered to change the interview questions. For example, question 4 was rephrased to; How does your organization handle customer data across various systems and touchpoints? Question 7 was rephrased to; Can you give an example of how customer insights attained from MDM and SOA influenced a recent business decision? Questions 9 and 10 were swapped to have a better- flowing interview. The changed interview questions were accessed when the field study participants approved, upon which the IRB application was then submitted. The field study assured that the interview protocol would help in capturing the information needed to address the research questions.

Credibility and Transferability

In a qualitative research study, the researcher must properly prepare the research design and identify individuals who will participate and share information from their perspective. Ensuring biases are addressed when necessary and the sample is a representation of the population. Salmons (2016) noted a researcher familiar with the work and processes may find it easier to recruit participants but must be cautious of potential biases. While familiarity can enhance understanding, it may also affect participant trust and data transparency. Conversely, an external researcher can offer an impartial perspective, which might be perceived as more credible. To mitigate biases, the

researcher carefully crafted questions based on literature and participants' experiences. It was imperative to elucidate the interview questions, and the corresponding responses provided by participants, particularly in relation to applying a qualitative study and assessing the validity.

Sampling a representative group of IT professionals to ensure they see the value in participating was crucial for effective data collection. According to Maite Ferrando et al. (2019), in the context of qualitative research, transferability can inform the creation of policies and procedures across various digital platforms when applied appropriately. Through transferability, there is a sense of consistency, and the researcher can create situations that can test trustworthiness (Erlandson et al., 1993). This was completed during the study, after the data collection was completed and documented. To support this process, the researcher ensured the information collected was from a credible source, and these research techniques can also be applied and replicated on a larger scale. Therefore, having a group of participants who can give their perspective on a retail business model and how information data systems can support or fail if implemented incorrectly can be valuable. The techniques discussed above were considered and expanded upon throughout the data collection and data analysis processes. Their contribution to the field of information technology should also be addressed and emphasized.

Data Collection

Preparation for data collection started two weeks before the interviews. Recruitment was conducted through business email groups and participants were contacted to obtain organizational authorization. Salmons (2016) emphasizes the

importance of obtaining participants' consent. The recruited participants were sent the informed consent via email and were expected to send back the signed informed consent to the researcher within one week. The recruitment process was conducted for two weeks. Interviews were scheduled according to participants' availability and conducted online via Zoom due to COVID-19. The estimated duration of the interviews was 45 minutes. However, each interview lasted 45 to 60 minutes based on factors when there were internet connection issues, or the relevance of the information provided by the interviewee.

In this research, interview questions were designed to capture each participant's experience and forced the participants to reflect and assess various business processes and information technology decisions. Manser and Mitchell (2012) highlighted interviews as a well-established qualitative method for assessing IT systems in the retail industry. Through this data collection tool, the researcher gained access to information, and data will be analyzed and synthesized. Using open-ended questions with follow-ups to ensure comprehensive responses and avoid missing information. Responses were transcribed verbatim and analyzed using content analysis to identify common themes. The analysis focused on responses to open-ended interview questions based on participants' knowledge and experience.

Data Analysis

Manser and Mitchell (2012) suggested an effective strategy to analyze the research data and interpret the results by identifying categories or chunking phrases that are similar. The goal is for the researchers to highlight some of the key findings that will answer the overarching research questions. By analyzing this data and synthesizing it into

categories of effective processes and data systems, professional technology practitioners will help inform the retail business industry and the field of technology. The data needed to be reviewed multiple times, and no information should be omitted. This helped to ensure the credibility and validity of the data and ensure the data findings included all participant responses. The researcher analyzed data using the content data analysis strategy. This strategy included describing, summarizing, or showing data points constructively to discover patterns that fulfilled every condition of the data gathered in the research. The following steps were taken based on thematic analysis:

1. Familiarization with the Data: Immersing in the data by reading and re-reading transcripts to understand the content and note initial ideas. This helps identify meaningful patterns (Braun & Clarke, 2006).
2. Generating Initial Codes: Develop initial codes by tagging significant data segments with labels representing key concepts. This iterative process involves breaking down the data into manageable chunks, often generating codes inductively (Saldaña, 2016)
3. Searching for Themes: Group codes into potential themes by collating them into broader categories representing overarching patterns. Refine themes through repeated examination of the data to ensure they capture the data's essence (Braun & Clarke, 2006).
4. Reviewing Themes: Assess and refine themes to ensure they consistently and coherently represent the data. Adjust themes by splitting, combining, or redefining them as needed (Clarke & Braun, 2017).

5. Defining and Naming Themes: Clearly define and name each theme, providing concise descriptions capture their core essence and relevance to the research questions (Braun & Clarke, 2006).
6. Writing Up: Present the themes with supporting data excerpts, interpreting how they answer the research questions and contributing to the broader understanding of the topic. Integrate thematic findings with the research context for a comprehensive and accessible analysis (Braun & Clarke, 2006).

The findings in this section highlighted the pivotal role that Master Data Management (MDM) and Service- Oriented Architecture (SOA) frameworks play in optimizing business outcomes for the retail industry. By improving data governance, systems enable business to maintain high-quality data, which enhances decision-making processes across various levels of the organization. Efficiently integrating customer information through MDM ensures businesses can meet customer expectations, leading to increased satisfaction and loyalty. SOA helps streamline processes, making data accessible and usable for different departments. This has led to significant improvements in employee productivity and operational efficiency. However, it is crucial to address the remaining limitations, such as the complexity of integrating disparate data sources, to capitalize on the potential of these frameworks. The insights gained from this research point to the need for a structured approach in future MDM and SOA implementations to ensure their long-term success.

Chapter Summary

Chapter 3 thoroughly summarized the research design and design appropriateness, research questions and instrumentation, population and sample, consent, confidentiality,

data collection, and data analysis process. Through each process, the importance of using action research will be discussed to better understand the collaboration of technology professionals to improve processes, implementation of evidence-based systems, interventions, and evaluation of data management systems.

Researchers must reflect on the conceptual framework identified for the study and ensure any emerging data or information is used to develop or inform new ideas or theories (Manser & Mitchell, 2012). Based on the information shared on the course room tools and dissertation guidance, the qualitative data analysis will be the basis of chapter 4 of the research study. This is where the research provides a comprehensive assessment and describes the research findings. Chapter 4 will analyze data collected using open-ended interview questions. The researcher will synthesize this data to inform the information technology field of study and the retail business.

Chapter 4

Analysis and Results

This qualitative study reviewed the implementation of Master Data Management and Service Oriented Architecture in the retail industry. The study aimed to determine the role of effective data management systems and sound event-driven architecture in measuring and acquiring business insights, the impact of MDM and SOA on the quality and synergy of data across enterprises within the retail industry for the customer domain and the requirements enterprise-wide data solution should have in the retail industry. Chapter 4 provides the research questions, discussing the data collection process, demographics, and data analysis before presenting a detailed discussion of the research results.

Research Questions

Research questions inquire about relevant aspects of the research topic. Answering the questions could help achieve the study's objectives. The three research questions adopted for this study are as follows:

R1. What is the role of effective data management systems and a sound event-driven architecture in contributing to the process of measuring and acquiring business insights? This question is explored through sub-questions focusing on data management, governance, implementation, and analysis.

R2. What is the impact of MDM and SOA on creating process improvement in quality and synergy of data across enterprises within the retail industry for the customer domain? This question is addressed through sub-questions examining process improvements and the synergy achieved through these frameworks.

R3. What are the impacts of implementing enterprise-wide data solutions have in the retail industry? Sub-questions for this research question delve into broader organizational impacts of data solutions, including strategic decision-making and customer satisfaction.

Data Collection

The participants signed an informed consent form sent via email as a pre-condition for participation. The informed consent form was signed electronically using DocuSign to ensure participants have agreed to the terms and understand their role in the study and their rights were protected at all costs. The form included various components mainly an introduction to the study, the purpose of the research, types of research intervention, participant selection, voluntary participation, research procedures, research duration, risks, benefits, reimbursements, confidentiality, the procedure to be used in sharing research results, right to withdraw or refuse and who to contact during the research. The participants conducted virtual interviews, which granted both full potential of personal information and freedom to express themselves. All interview sessions spanned between 45 to 60 minutes in duration. The recorded interview data obtained consent before being properly transcribed and applied for analytical purposes.

Demographics

Twenty technical and business experts participated in this research. The participants provided insights primarily based on their experiences within their respective companies, offering limited perspectives on the broader retail industry. The demographics of the participants are indicated in Table 4 and Table 5 below.

Table 4*Sample Demographics*

Gender	Number of participants
Female	8
Male	12

Table 5*Experience Demographics*

Years of experience	Number of participants
1-3 years	3
4-6 years	6
6-10 years	8
Over 10 years	3

Data Analysis

Data analysis is a fundamental component of the research processes, ensuring that findings are systematically organized, interpreted, and presented to answer research questions effectively. Data analysis allowed the researcher to transform raw data into meaningful insights, facilitating the identification of patterns, relationships, and themes that address the study's objectives. Data analysis helped to draw conclusions and enhance the credibility and validity of the study. This research adopted a combination of narrative and grounded theory approaches for data analysis, both recognized for their rigor and ability to in-depth insights into complex phenomena. The grounded theory approach was particularly valuable in this study, focusing on generating a theory grounded in the collection data. This iterative process involved constant comparison of data, coding, and categorization to uncover underlying patterns and themes. By employing this strategy, the researcher was able to construct a conceptual understanding of the role of Service-Oriented Architecture (SOA) and Master Data Management (MDM).

In addition, narrative analysis was used to explore the perspectives and opinions of participants, providing a rich qualitative dimension to the findings. Through this method, the researcher examined participant accounts to identify insights related to the research questions, such as the impact of SOA and MDM on productivity, customer satisfaction, and operational efficiency. As Nowell et al. (2017) explained that narrative analysis is particularly effective for capturing the lived experiences and subjective views of participants, allowing for a nuanced interpretation of the data. By integrating grounded theory and narrative analysis, the researcher ensured a comprehensive approach to data analysis. This dual methodology allowed for both theoretical development and detailed exploration of participant perspectives, ensuring that the findings were robust and aligned with the study's objectives. This rigorous analysis process ultimately contributed to answering the research questions and advancing knowledge on the implementation of SOA and MDM in the retail industry.

When considering the theoretical framework of, the three framework theories were applied. First, the diffusion of innovation theory guided the analysis by helping identify stages of adoption, from awareness to implementation. Second, resource-based theory focused the analysis on how MDM and SOA serve as strategic assets to improve data quality and performance. Lastly, the dynamic capabilities theory helped assess how businesses adapt and configure their information technology (IT) systems over time in response to change.

Results

This section introduced the challenges and opportunities that retail businesses encounter when implementing Master Data Management (MDM) and Service-Oriented

Architecture (SOA) systems. MDM and SOA offer significant benefits by improving the integration and governance of data across different business units, enabling better decision-making and operational efficiency. The discussion explored how these frameworks enhance customer information management and how their effective implementation can lead to improved customer satisfaction and retention. However, the section also addressed common obstacles, such as difficulties in integrating data across departments and ensuring data security privacy. It is essential to consider the best practices for overcoming these challenges and ensuring the success of MDM and SOA frameworks. The section highlighted the importance of leadership in driving the adoption of these systems and processes within retail businesses. Finally, the upcoming discussions provided insights into the strategies for mitigating potential risks associated with data management.

Data Management in the Retail Industry

The business and technical experts were asked about the data management process in the retail industry based on their experiences working in different retail companies. Participants stated that the data management departments include IT, Supply Chain, Finance, and Purchasing. The business experts noted that the data management role is mostly carried out by the department that initiated it or developed the database systems being used. Two business experts noted that there is a need for a separate division to oversee the storage, processing, and security of data. One business expert stated, “Most companies in the retail sector prefer a general data management team for developing, controlling, and securing the company’s digital infrastructure and database.” The team functioned independently to ensure the requirements and wishes of other

departments in the organization are met. Participant 2 stated “an ideal data management framework would include placing a master data manager between an individual department and a general database.” This is to ensure the MDM works as an intermediary in charge of storing and processing data generated by operational and strategic activities and processes into a company’s database to develop a high-quality database. They stated a specialized MDM should have specific knowledge based on the department in which it is being implemented. The business and technical experts were asked about the skills and competencies needed to organize various data management components. The technical experts stated computer literacy is the most important skill and competence. However, business experts have noted computer literacy does not bring value to data management because department-specific skills are also applied in data management.

Integrating MDM and SOA

Business architectures stated most of their customers request the integration of MDM and SOA to efficiently manage data relating to business insights. Technical experts noted the integration of MDM and SOA expands the information gathered, thus enabling the organization to make necessary changes to increase customer satisfaction, sales, and employee productivity. Information gathered and stored includes customer-related information such as customer preferences and changes in customer behavior. The study participants were asked to identify the types of information regularly gathered by the companies they have worked with through the event-driven architecture combines SOA and MDM. The information was gathered through interview questions that were open-ended. For example, Participant 12 commented that “data evolves. Our systems must evolve with it,” which highlights the need for flexibility in lifecycle management.

The main customer-related data options research participants had to choose from were organizational, market, individual, and business unit levels. Some data can be collected at more than one level. For instance, data on customer satisfaction can be gathered at the business unit and organizational level. Tables 6, 7, 8, and 9 below summarize responses.

Table 6

Market Level Customer Information

Customer Information collected and stored	Respondents	Percent of companies use MDM and SOA to collect the specified customer information
Customer's business environment	10	55%
Customer's market	10	45%
Customers' interaction with other customers	10	20%

Table 7

Organizational Level Customer Information

Customer Information collected and stored	Respondents	Percent of companies use MDM and SOA to collect the specified customer information
Sales history	10	100%
Contact information	10	50%
Customer feedback	10	75%
Customer satisfaction	10	55%

Table 8

Individual Buyer-Level Customer Information

Customer Information collected and stored	Respondents	Percent of companies use MDM and SOA to collect the specified customer information
Information on buyers	10	100%

Table 9*Business Level Customer Information*

Customer Information collected and stored	Respondents	Percent of companies use MDM and SOA to collect the specified customer information
Correspondence evaluation	10	58%
Buyer top management	10	65%
Customer support	10	75%

Integrating customer data through Master Data Management (MDM) and Service-Oriented Architecture (SOA) frameworks was essential for improving customer satisfaction, as it allows for a more streamlined and consistent approach to managing business data. These frameworks ensure that accurate, reliable, and actionable insights are generated, which enables businesses to make better informed decisions across different functions. Additionally, these frameworks contribute to improved data consistency and governance, particularly when managing large databases across multiple systems. The ability to synchronize customer data with other operational data enhances overall business performance, leading to a more efficient and responsive business model. Participants in this study echoed these sentiments, confirming their data management practices have improved significantly since adopting MDM and SOA. The frameworks have empowered organizations to better predict customer needs, allowing them to tailor services more effectively. This alignment between data and business goals is critical for long-term success in the highly competitive retail industry.

The data in the tables above indicated many companies (55%) use SOA and MDM to understand the customer's business environment. Participant 9 stated "companies need to understand their potential customers' needs and preferences to create

a competitive advantage by offering goods and services that meet these needs. Therefore, the company needs to gather insights on factors influencing customer needs and preferences, such as price and quality.”

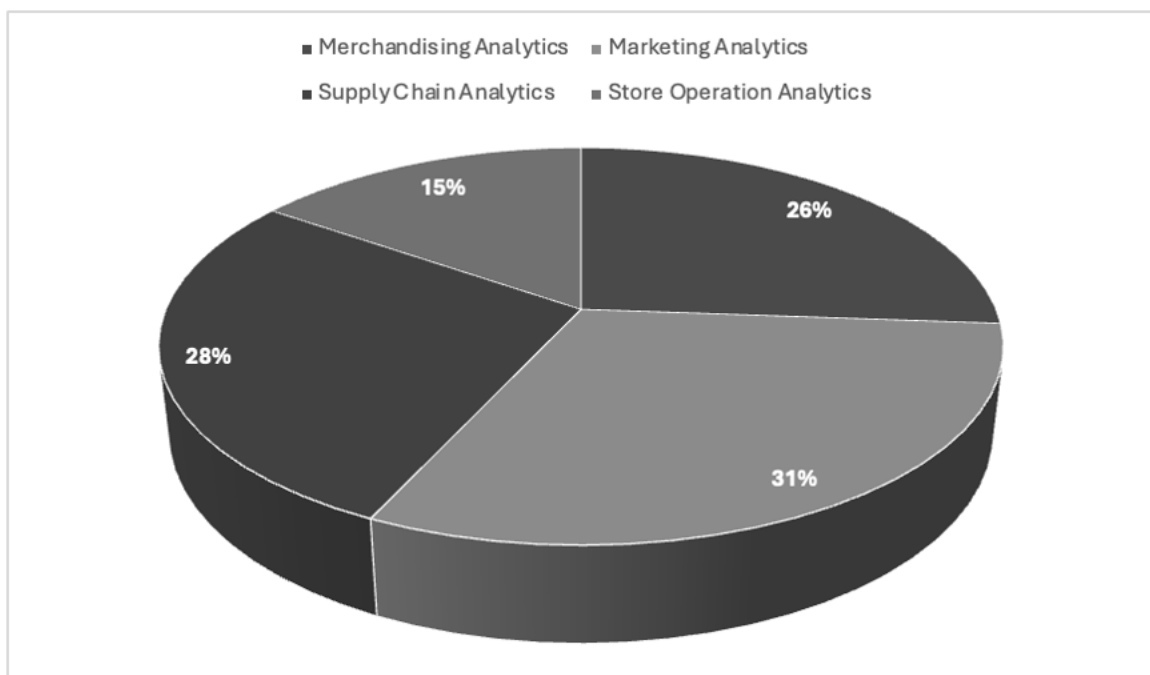
Participant 4 stated “price may also influence customer perception of a product. For instance, some customers perceive high prices as an indicator of high quality, thus dictating purchase decisions.” The data also indicated many companies (45%) use SOA and MDM to understand the customer market based on the distribution of customers in different regions. Few companies (20%) use SOA and MDM to understand customer interactions with other customers, hence determining how customers may be influencing sales through referrals. This information is essential in designing a marketing strategy because companies may opt to use referrals to increase product awareness. The data also indicated most companies (100%) companies use SOA and MDM to understand sales history to determine the company’s success. Some companies (50%) use SOA and MDM to get customers’ contact information. The contact information may be used for direct marketing and acquiring customer feedback. Many companies (75%) also use SOA and MDM to get customer feedback on the products and services offered. Feedback may focus on product quality. Most companies (55%) also use SOA and MDM to gather information on customer satisfaction. Research respondents also stated companies (58%) use SOA and MDM to evaluate correspondence with customers. This information is crucial because it helps the organization determine whether the company’s customer engagement strategy is effective. Most companies (75%) also use SOA and MDM to gather information on customer support which is also considered in evaluating customer

relationship management. Most companies (100%) also use SOA and MDM to get information on the distribution of buyers.

Research participants were also asked about the common types of business analytics used in the retail industry. The business analytics included merchandising, marketing, supply chain, and store operations. The results are in Figure 10 below.

Table 10

Retail Industry Business Analytics



The chart indicated most companies in the retail industry also use supply chain analytics. One of the business experts stated, “most companies in the retail sector are implementing GPS tracking to monitor the movement of goods.” Another business expert stated, “business experts are applying advanced supply chain analytics to understand the behavioral patterns of drivers. They also identify risky points on a driver’s routes and minimize unnecessary stops, hence enhancing the safety and efficiency of logistics.”

Participant 3 stated,

Retail companies use merchandising analytics to align merchandising decisions with client expectations through effective assortment planning, space allocation, and product adjacency. Retailers can identify customers' preferences for every stock-keeping unit and use the preferences to forecast demand while discovering new products may attract more customers.

Further, Participant 3 explained, “few retail companies use store operations analytics, which includes predictive analytics. Business experts noted most companies in the retail sector use sensors to measure inventory levels and automatically restock their shelves.” Four technical experts labeled as Participants 2, 3, 4, and 5 stated the future application of store operation analytics will include the application of the Internet of Things (IoT) to create smart price tags that can automatically change prices.

Big Data

The business and technical experts provided insight on the application of Big Data in the retail sector. Ninety Five percent of the research participants stated that they use Big Data in developing customer loyalty by personalizing offers. Eighty percent of the research participants stated that they use Big Data to target the right customers. In addition, 65% of the research participants stated that they use Big Data to make better offers and 25% stated that they use Big Data to improve segmentation. The responses are summarized in Table 11 below.

Table 11*Implementation of Big Data in the Retail Industry*

Application	Number of Respondents
Developing customer loyalty by personalizing offers	95%
Target the right customers	80%
Make better offers	65%
Improve segmentation	25%

The business and technical experts explained the circumstances under which customer data should be shared with third parties. Seventy percent agreed that retail companies should share customer data with third parties at all times while 30 % disagreed. Eighty percent disagreed with retail companies sharing customer data with parties only if it gives value, while twenty percent agreed. Ninety-five percent agreed that retail companies should only share data with third parties on a contract basis while five percent disagreed. Lastly, eighty percent agreed that all customer data should be destroyed after use while twenty percent disagreed. These responses are summarized in Table 12 below.

Table 12*Sharing Customer Data with Third Parties.*

Statement	Yes	No
Retail companies should share customer data with third parties at all times	30%	70%
Retail companies should share customer data with third parties only if it gives value	80%	20%
Retail companies should only share data with third parties on a contract basis	95%	5%
All customer data should be destroyed after use	80%	20%

The business and technical experts explained how Big Data collection is used in the retail sector. The research participants stated that big data could help develop customer loyalty by making it easier to distinguish between customers and develop

customer segments by understanding customer needs to make personalized offers. Their responses are outlined in Table 13 below.

Table 13

Big Data Collection and Use in the Retail Sector

Statement	Yes	No
Big Data can help in developing customer loyalty	100%	0%
Companies should outsource Big Data	80%	20%
Companies should perform Big Data Analytics themselves	75%	25%

Challenges in the Implementation of MDM and SOA

The business and technical experts were asked about some of the major challenges encountered in implementing SOA and MDM in the retail industry. Five technical experts stated that one of the challenges is the difficulty of reading and inserting data into the IFS database because different suppliers have various definitions of a product. All the business and technical experts also stated that the cost of developing solutions for reading data about a product in various ways for different suppliers is a major problem that most retailers have yet to eliminate. Two technical experts added that data maintenance is also a major challenge for many retailers, as it becomes problematic with an increasing customer base. One technical expert labeled as Participant 15 stated that a “major challenge connected to supplier and customer data is building a stable data structure.” Ten business and technical experts also identified access to data as another factor creating challenges because it is difficult to find a balance between access control and transparency. Participant 6 stated, “master data lifecycle management is vital, but because of the large product data entered an ERP system, it becomes hard to manage the data.”

Participant 11 argued that the biggest challenge for IFS integrated into MDM is related to scalability, whereby the structure of all customer data is first built with a strategy an organization operates at a single physical location. Participant 9 stated that a major challenge of master data management in the retail sector is that customers do not always comprehend the significance of managing master data. Still, even if they comprehend, they may not have enough resources and time for structuring and cleaning their master data. The business experts suggested it is vital to clean master data before it is transferred into the updated or new system when updating or implementing a system within a retail organization.

Five business and technical experts stated in understanding the needs of managing master data, it is important to understand the complexity of the setting where the data is managed. One of the technical experts stated, “understanding how master data is integrated between systems in a system setting is also essential. Product data is often developed in a Product Data Management system where the product is designed before being forwarded to an ERP system.”

Two technical experts labeled as Participants 3 and 14 stated that different types of possible integrations are required between ERP systems and other systems implemented in organizations. Four business experts also stated customers are not mature enough to manage different integrations, thus initiating a debate on data ownership. One business expert stated it is important to agree on who is responsible for data management and how the value of master data is determined. Three technical experts stated master data integration is a challenge in lifecycle management. One technical expert, Participant 17 stated,

It may be hard to remove product data in a system because of the integrations connected to the product from other systems. It may also be hard to track where a product has been removed and where it exists in a system setting, including many sub-systems and interactions.

Participant 8 stated that organizations within the retail sector have many integrations because the same information is usually used on various channels. Two business experts stated there are standard integrations to manage integrations between different systems. They added integrations that require adjusting based on customer needs. Five technical experts stated using different systems and channels to show the same information is important. In addition, another technical expert explained there are no guarantees that the integrations will work perfectly. Four technical experts stated changes might lead to integration bottlenecks if a change is applied to a lot of product data used within various systems. Two business experts stated customers set the integrations themselves, and the integration should work with an interface that automatically facilitates integrations to other systems.

Ten business and technical experts stated managing master data is important throughout the entire data lifecycle, although there are possible challenges associated with different lifecycle stages. One of the business experts stated customer data could be used in multiple systems during data lifecycle management in a supply chain planning system and Customer Relationship Management in a financial and ERP system. Two business experts stated significant product data should be managed in the retail industry, which changes with time because of the releases of new collections at different times of year. Five business experts stated removing and phasing out data could be challenging

because of multiple sources referring to similar data. Five technical experts stated system support offering an overview and facilitating progress is needed. The technical experts also stated data maintenance might be facilitated by providing interfaces for data storage related to specific master data. Ten business experts stated product attributes could be used to set the status of a product to inactive or active to manage product data during the data lifecycle or to decide if a product should be exported to other systems rather than an ERP system. Four technical experts stated if master data is removed, there may be performance issues when the data constantly increases, particularly for organizations within the retail sector due to the intensive flow of product data in the sector.

Framework Development

Three business and technical experts stated the implementation of master data management and service-oriented architecture in the retail industry begins with engaging in Big Data optimization to use Big Data to improve retail operations. Participant 16 identified the first step in applying for a master data management and service-oriented architecture is collecting data from merchants. The business expert can be collected through payment data, loyalty systems, external resources, and merchant data. The external sources include assessments of factors that influence purchase behavior. The business and technical experts stated the data collection process in the architecture incorporates comparing data from different sources. Business and technical experts stated in this phase, retail companies transform the data analysis output into insights can be used to optimize in-store customer experience or create an effective customer loyalty program. They stated Big Data analysis might be conducted without a specific goal, and it is later

determined how the insights can be used. Insights could also be real-time when the right software for real-time data analysis and insights is available to the retail company.

Participants 4, 18, and 19 also stated getting good insights from Big Data analysis requires using specialized people such as data experts and scientists. Business experts also stated customers might use the hard and soft benefits offered to them and spend them in the companies' stores. They stated more loyal customers spend more time visiting stores that offer good benefits and order extra items. One business expert labeled as Participant 7 stated,

Payment providers in the retail industry also use data analytics to make a comparison within the retail industry. Merchants benchmark to decide on the changes need to be made and begin improving processes for either increasing revenues or reducing costs.

Three business and technical experts stated the architecture must deal with privacy regulations are vital in all customer data processes. Participant 12 stated, "companies need to observe government regulations. Another important aspect that needs to be put into consideration is data ownership." The business and technical experts stated, when a retail merchant wants to outsource data analytics or intends to work with a third party, they need to closely review agreements because the merchant owns the customer data and is responsible for upholding customer privacy.

The technical expert mentioned the output, input, and speed of deriving customer data are different in Big Data and normal data. One business expert stated data collection includes merchant data and data sources. This increases the complexity of data analysis, thus, the need for new software solutions. Three business and technical experts stated

merchants are in the early stage of determining whether the new software can meet the organization's needs. They added the insights gathered from the analysis become faster and more specific in real-time. Participant 19 stated "the decisions on customer loyalty options also change when merchants use Big Data. Customer loyalty development options become more personal, thus enabling the company to target specific customer needs and increase sales."

Requirement Analysis

Participants 1, 13, and 20 stated the requirements of the MDM and SOA architecture need to be analyzed to enhance efficiency. They stated after various requirements have been identified, a requirement analysis should be conducted to sort out any irrelevant requirements and look for potential conflicts or overlaps. They stated the analysis could be used to prioritize the final requirements in the requirements specifications. Participant 2 stated defining system boundaries in the first step of the requirements analysis is crucial to determine the requirements are not within the architecture's scope. Another technical expert suggested constructing and using a checklist to perform a requirements analysis systematically. The checklist aims to find the problems in the identified requirements. The questions in the checklist should be designed based on the specific attributes of the desired system. One of the technical experts suggested using software to support the negotiations regarding requirements and to plan for managing conflicts.

The second step in requirement analysis was identified as the prioritization of requirements to define their importance in creating a successful system and their importance to stakeholders. Participant 18 recommended using the priority classification

with various categories to choose from and separating the most vital requirements from the desirable ones. Another technical expert proposed using a multi-dimensional approach to classify requirements to find similarities and overlaps in the requirements. Therefore, it may be possible to combine various requirements and find missing requirements in various classification dimensions. The technical experts suggested not using over six different classifications to decrease the risk of having too limited dimensions. The technical experts proposed using keywords such as database, user interface, system, and security and communications when classifying requirements. They stated when all requirements have been classified based on different dimensions, comparing requirements within similar dimensions is possible. The technical experts recommended using matrices to compare the different requirements to identify overlaps and conflicts. They added a risk analysis is needed to identify different risks connected to the implementation process.

Participants 2, 5, and 16 stated after the requirement analysis, a specification of the requirements should be written. They stated one of the challenges encountered when writing down the final requirements is they may be written vaguely and be expressed as unclear ideas during the requirements interpretation phase. They recommended investing more time and effort in writing the descriptions to increase the likelihood of achieving the required qualities because it can be complicated to write high-quality requirements. Three business experts stated those writing framework requirements should consider the requirements may be read by individuals with different backgrounds and diverse knowledge connected to the included requirements. They recommended using standard templates to enhance the consistency of the descriptions, concise language, and short

sentences. The technical experts stated descriptions should include statistical formulas or programming language rather than natural language if using the formulas increases the clarity of the requirements being described. They also stated diagrams could be useful in demonstrating relationships between the requirements.

Factors Influencing Data Management Model Selection

Participants 3, 9, and 11 indicated that data quality is one of the factors influencing data management model selection. One of the business experts noted the time spent on manual data correction affects the finance and IT departments. The expert added the finance and IT departments are more involved in data quality because they use data for decision-making daily. Another business expert stated that the business segment is more concerned about the speed at which data is delivered because they use the data after it has been corrected. Three business experts also stated reporting data for the finance and business segment is important, particularly for finance, because of the direct connection to reporting and primarily because of an organization's commercial nature. Three technical experts stated customers are affected by the quality of data. The technical experts also stated the IT department is responsible for dictating how data is transmitted while the finance department contributes to eliminating the issues created by the complexity of different systems. The technical experts added that domain issues arise from collection and delivery because the business domain is directly connected to the end-user and their fast response to a customer request is vital.

Three business and technical experts identified the second factor influencing the selection of a data management model as the internal perception of Master Data Management. The technical experts stated retailers assume MDM resolves all data

management issues, but assumption is wrong. They stated that the business domain has varying ideas related to data management challenges of future digitalization of the organization. Five business experts stated retailers view MDM as a technological tool for improving how businesses operate. They added the IT domain is clear on MDM, and there has been an improvement in how it is handled in the retail sector. Three technical experts also stated the perception of the business domain is the same as the IT domain, and some architectural schemas are considered. Two business experts noted the finance department has a diverse perception of MDM due to the financial nature of the data used in the department. They added finance experts emphasize the master data owner should be from the finance department.

Three business and technical experts stated previous projects related to the management of master data influenced the model selection. Two technical experts noted the IT department is mostly involved in various projects aimed at guaranteeing the reliability of ongoing projects. They added many initiatives have eliminated the challenges connected to master data. They reported the finance department provides the right quality, formats, and definitions of data to improve the efficiency of master data management. Three business experts stated the impact of data quality on decision-making plays a significant role in determining the master data model selection. One technical expert stated global standardization and local flexibility also affect model selection. In addition, the business experts stated the balance between global standardization and local flexibility is important and highly complex. The technical expert stated flexibility is part of countries' commercial nature and ability to carry out business operations. The business experts stated from a business viewpoint, finding the balance between local flexibility

and global standardization is important because a company needs to stay updated on local needs. They added centralization may reduce customer relationships.

Two business experts stated internal culture also plays a vital role in model selection. The business experts added the finance domain is also connected to the flexibility of doing business on a country level because of the daily data checking. Three technical experts stated the difference in data definitions also influences model selection. They stated the IT and finance departments have a broader understanding of data than the business domain because of different definitions by systems and departments. Three business experts stated the finance domain often deals with discrepancies in the definition of data while the IT department relies on the finance definitions for various terms and data ownership. One business expert stated the business and finance domain has a positive view toward MDM implementation, although the IT department encourages strict implementation due to the legacy systems put in place and their complexity.

Architecture

Three business and technical experts stated the implementation of MDM and SOA architecture in the retail industry requires an architecture facilitates a more efficient way of managing master data in terms of data maintenance, data implementation, performing data modifications, and searching for specific data. Two technical experts recommended building a master data component where customers can buy the MDM component when they apply IFS solutions. Four business experts also noted managing an environment, including the integration and co-existence of multiple systems, is one of the success factors of the MDM component. Two technical experts indicated retailers complain it is hard to get a product overview, and the characteristics connected to it.

They also stated it may seem unnecessary for some retailers to manage similar data types of multiple times because of the product structure. One of the technical experts stated the challenge of creating a product where the product and the characteristics can be managed in one place and collecting information associated with the product is the most important challenge to address for the consulting sector. Another technical expert gave an example when developing a stock or sales product with similar characteristics, instead of developing a new product for every site where the product is used, characteristics specific to a site should be implemented as the site's exceptions. The expert stated this logic would still lead to fewer exceptions and modifications. Another technical expert suggested using a site-buddy to efficiently manage data implementation where a site can be a reference when adding data to a site with the same attributes and properties. The expert added it would be technical to work on changing the structure.

One business expert stated a more central product data management may result in challenges associated with data access but might solve current maintenance challenges. The expert added the ability to offer information connected to a product is valuable, which is why the attribute is added to the resulting requirement. Two technical experts suggested developing data management structures by focusing on retail assortment and adding profiles connecting products instead of storing all information about a product on the product level. They stated it would be valuable to collect information in a profile connected to a product to sort products with similar information by a specific profile and later perform changes for various sites simultaneously. One business expert stated managing products across different sites is the most vital perspective in a future MDM architecture. Two technical experts stated focusing on stock-keeping units may be

problematic due to the need for changes in the product line; hence, the need for catalogs to collect product data, which would be easier to move a product catalog to a new site if the need arises.

Availability and Integration

Four business and technical experts stated master data should be integrated within a system environment with various co-existing systems. The business experts stated even though the management of master data is discussed for data within the same database, few companies are using one database to store all data. Two technical experts stated to avoid the bottlenecks in integration may happen when master data changes are performed, a user should preview the impact a change would lead to, thus decreasing the risk of conducting changes with unanticipated results in subsystems. They suggested providing a more adaptable solution to integrate data between IFS excel and applications to migrate data efficiently. Three technical experts recommended using a data migration tool but stated it requires technical skills. They stated it would be better to use product families from suppliers without adding the data in an ERP system so all products in the supplier assortment can be viewed and searched without creating the products in the IFS application. Two business experts suggested providing product catalogs to customers where the supplier manages system updates to ensure the customer can access the latest product catalog. Three technical experts recommended using global standards when designing the integration of master data.

Governance

Three business experts stated the involved stakeholders must establish processes and rules to define details of how master data management should be done. They stated

managing master data requires facilitating decisions on the ownership of data and integrating systems. They added MDM needs to be defined better in terms of what is included and set up rules on sharing data between sites. Technical experts stated the risk of errors increases if data is copied without regulations.

User Interface

Three technical experts stated the MDM user should be given more advanced possibilities for data visualization. They reported retailers should invest in visualization prospects to view how different products are connected and analyze the consequences of making various changes. Four business experts stated customers prefer more usability when using an ERP system. One of the business experts said retailers creating orders using an ERP system usually have technical knowledge. Still, it may be hard for new consultants to learn about the management of the system. One technical expert stated customer projects need large system configurations. Another technical expert stated

implementing MDM in data management is efficient, powerful, and flexible because it is possible to store various attributes of a specific product. However, retailers using MDM require programming skills and knowledge about the administration process. Three technical experts stated customer organizations rely on IFS to perform updates and small alterations in the system. Two business experts noted the tools for customer data analysis would be essential in finding the patterns in customer behavior. They added more product requirements are important within e-business to sort products based on their attributes. Three technical experts stated there should be rules for regulating various attributes in large markets. They noted future MDM architecture should include product history to keep track of the changes in the products and suppliers.

Lifecycle Management

Three technical experts suggested creating a process of phasing out product data and implementing new data in the ERP system. They stated system support is required to manage the two processes. Two business experts stated companies would benefit from monitoring product data to eliminate irrelevant data and ensure no data is duplicated. They stated monitoring data implemented from integrated systems, particularly data from companies with large suppliers, is critical to ensure agreements are kept about processes and products. One technical expert stated the data implementation process should be automated to ensure different users need to edit and approve the implementation before it can be applied within the system setting. Three business experts stated the support of inserting product data within the retail business is vital because there is a need to manage a large amount of data. Two technical experts reported managing complex data phasing out processes requires providing system support for product data management when the product is in stock, when it should no longer be purchased and when it can be removed from the systems.

Aligning SOA and MDM

Three technical experts indicated organizational culture affects the alignment of SOA and MDM. They gave an example when IT and businesses operate closely, the development of SOA and MDM is better aligned with IT objectives than in organizations where the collaboration between business and IT is not part of the organization's culture. Two business experts stated business unit-oriented organizational structures have an inhibiting impact on the realization of the alignment of business and IT in SOA and MDM development. They said business silos promoted a culture where individuals have

to take ownership of specific processes, data, and IT systems. They added in such situations, data sharing, services, and open interfaces are not accepted because of the fear of losing influence and power within the organization. Three technical experts stated the skills needed in the SOA and MDM design phases, such as modeling business processes and service design, are important in SOA and MDM architecture development. Two technical experts noted SOA and MDM development should focus on understanding processes and services. They identified service design as one of the most challenging parts of MDM and SOA development. Further, three business experts stated the participation of business stakeholders is vital in MDM and SOA development. They indicated stakeholder participation in MDM and SOA development positively impacts the alignment of the developed architecture with organization objectives.

Implementation Factors

Three business and technical experts stated technical support is one of the most important factors in implementing the MDM and SOA architecture in the retail industry. They specified technical support influences the successful integration of software and implementation of new elements in an MDM and SOA framework. One technical expert stated the outcome and process of implementing MDM and SOA depend on existing data architecture and systems. Another technical expert noted the compatibility of MDM and SOA software solutions and existing systems requires to be compared and evaluated to determine whether existing systems have been altered, or the targeted MDM and SOA software gets interoperability. Three business experts stated good communication, including IT and other departments in SOA and MDM implementation, is needed for the company to benefit from technical expertise and find preferences are more likely to be

accepted. One technical expert stated the current situation of existing infrastructure should be evaluated to identify possible solution options.

Three technical experts recommended implementing mechanisms for separating master data from other data and managing master data independently. They added retailers should use a data warehouse consisting of the full history of the changes made to the data. They stated the source of data warehouse entry lists is operational or transactional systems, and new data is generated when a transaction or event such as a customer purchase or a change in product inventory. Two business experts stated MDM ensures a unified view of key business assets. They indicated one of the main challenges in MDM without proper technical support is to achieve a balance in effectiveness, efficiency, and acceptance when modifying tools and systems. Three technical experts stated employees must be educated and included to enable change acceptance and willingness to follow defined standards. They added it is also important to involve all support groups such as IT, HR, procurement, and business administration.

Four technical and business experts identified management support as another implementing factor. They stated if the executives at the chief level of management understand the value and impact of MDM, management must make strategic goals and define goals. Three business experts stated the main strategic goal is whether MDM implementation should be conducted in a central, local, or hybrid approach. Two technical experts added processes and technologies need to be altered in a way supports the initial goal by making processes while personnel have enough time to adapt to altered processes and can use the technology to support their work. Three business experts stated the extent of implementing technologies and processes and the willingness of staff to

follow changes directly impact the data quality. They also noted an evaluation of the current general situation is needed before implementing MDM.

Four business experts stated data governance is an important implementation factor. They stated master data should be managed based on defined standards, including integrity and data security processes. Three technical experts stated successful data governance results in the quality of master data, which is complete, timely, and reliable. In addition, they added data security in data governance defines the data storage process, accessibility, and entry points to ensure the retention and privacy of master data. One technical expert stated integrated data enables an organization to use master data for decision-making and establishing a successful future of organizations. The technical expert stated a challenge faced in implementing data governance for MDM is joining, managing, and controlling all data sources distributed over various business units. Another technical expert said data governance naturally impacts the company's strategy and influences business processes because requirements and rules limit the speed and flexibility of a process. The technical expert recommended using a COBIT framework for IT governance to offer recommendations and guidelines on how to approach MDM implementation.

Three technical and business experts stated a major aspect of MDM is data ownership and the operational implementation of roles and responsibilities. They stated the main challenge related to MDM implementation is the clear distribution of roles and responsibilities. Two business experts stated the descriptions of existing roles should be adapted and overlooked if necessary. One business expert stated finding an owner for MDM and the implementation on the chief level of management can result in discussions.

This is because IT is only one of the many aspects of MDM implementation, and the Chief Information Officer might not have enough expertise for the task. Therefore, finding the right person to complete this role is complex. Another business expert stated assigning the MDM implementation role to a data steward result in some challenges because the responsibilities and tasks are hard to formulate. The technical expert stated the ongoing individual implementation of MDM in an organization must be observed and documented to effectively name tasks and responsibilities. One technical expert stated that because many parties are involved in MDM implementation, no one will take the risk of being held responsible for the data. The technical experts reported the role of a data steward in MDM implementation is vital because the responsibility for the quality of data and processes is vital for continuous MDM implementation. One business expert stated data quality issues are directly connected to processes, data management, and data creation, and these need close monitoring. One technical expert stated roles holding responsibilities for data stewardship are vital because their accountability for the quality of master data will guarantee the commitment of an enterprise to implementing MDM and SOA.

Themes on Desired Characteristics of MDM Architecture in the Retail Industry

The business and technical experts were asked about the most important attributes of their desired MDM architecture in the retail industry. Their responses were categorized into different themes. The analysis of the interviews with industry experts brought out several themes in the characteristics that they would desire for MDM architecture in the retail industry. The themes identified in the analysis have been explicitly linked to both the conceptual frameworks and the research questions to provide coherence.

Theme 1: Integration and Availability of Data

This theme addresses Research Question (R1) and was closely tied to Resource-based Theory (RBT). RBT underscores the importance of unique and inimitable resources, such as integrated data systems, in maintaining a competitive edge. Interviewees frequently described how their organizations utilized MDM and SOA to create and leverage such resources. For example, participants noted that centralized data systems improved operational efficiency by eliminating redundancies and ensuring consistent data quality. This supports the theory's assertion that rare and valuable resources lead to sustained competition.

Participants highlighted the transformative impact of MDM frameworks in consolidating disparate data sources, improving data governance, and enhancing overall data quality. Several participants noted that achieving seamless data integration was a critical step toward unlocking actionable business insights, emphasizing the value of MDM and SOA as strategic resources. Participants stressed how an effective MDM architecture should ensure easy flow of information to drive proper decision-making. One technical expert stated "integration is the spine of our data strategy. Without it, we cannot effectively leverage our data." Another technical expert stated, "we need a system allowing us real-time data availability across the platforms to remain competitive." One business expert stated "most of our decisions are based on integrated data. If it is siloed, we lose out. Another business expert stated data in the departments is critical to customer unified experiences."

Theme 2: Effective Governance Framework

The second major theme related to the effectiveness of the governance framework. This theme was aligned with the Diffusion of Innovation Theory and RQ2. This theme explored how SOA has been instrumental in streamlining workflows and enhancing data accessibility across organizational silos. Participants reported that SOA implementation significantly reduced process inefficiencies and enabled more effective cross-departmental collaboration. The diffusion of innovation theory provided a lens to interpret the adoption process of MDM and SOA within the retail sector. Participants' responses illustrated the phases of adoption, from initial awareness to implementation and confirmation, emphasizing the significance of relative advantage and compatibility. Retail organizations that adopted these technologies reported enhanced decision-making processes and better customer insights. These findings aligned with the theory's emphasis on how communication channels and innovation attributes and influence adoption rates.

Respondents indicated that for data to be of good quality, relevant, and compliant, the governance should be effective. One of the technical experts stated, a "strong governance framework holds integrity of the data and its security high—something that is unnegotiable in our line of business." Another technical expert stated, "without governance, there is a great risk of falling into data chaos. There is a particular need for data governance to uphold trust in the data." One business expert stated, "it's not about the rules, really, so much as ensuring our data serves the business well. Another business expert stated, "we need explicit policies to manage data quality and compliance. And governance lies at the base of all that."

Theme 3: User Interface usability

The third theme, user interface usability was linked to Dynamic Capabilities Theory and RQ 3 since the usability of the MDM system is a significant capability. The business and technical experts stated that a user-friendly interface should be the primary focus of the retail companies. One of the technical experts stated, “in case of complicated user interface, the rate of adoption will be low. It needs to be simple.” Another technical expert stated, “user-friendly design can make or break the success of our MDM implementation.” One business expert stated “employees will be using it, so it's got to be easy to navigate; it impacts productivity.” Another business expert stated, “the interface must be fit for the nontechnical user as well. Everybody should be able to access data with ease.”

Theme 4: Lifecycle Management

The fourth theme aligned with the Dynamic Capabilities Theory and RQ3 since proper lifestyle management is a capability. The business and technical experts stated that MDM systems should support the entire lifecycle of data right from its generation to archives. One of the technical experts stated, “if it is for managing data throughout its lifecycle, value will be maintained.” Another technical expert stated, “we need to ensure our MDM system can handle the data, from inception to retirement, effectively.” One of the business experts stated, “lifecycle management provides the ability to understand what data means and how it's used, not only today but over time, which provides you the ability to do proper strategic planning.” Another business expert stated “data is not static. It changes. And so should be our systems to manage that change.” Table 13 below

outlines the characteristics considered by business and technical experts when analyzing MDM and SOA.

Table 13

Characteristics of MDM Architecture in the Retail Industry

Characteristics	Number of respondents
Architecture	85%
Availability and Integration	80%
Governance	95%
User Interface	100%
Lifecycle Management	95%

Chapter Summary

The research findings in Chapter 4 have been aligned with the conceptual frameworks discussed earlier, including the Diffusion of Innovation Theory, Resource-based Theory, and Dynamic Capabilities Theory. This alignment ensured a comprehensive understanding of how these frameworks inform the results of the study. The adoption of MDM and SOA within the retail industry can be viewed through the lens of innovation diffusion. Stages of adoption, from knowledge acquisition to implementation and organizational success were highlighted in the results. Themes related to early adoption and organizational readiness underscore the theory's emphasis on compatibility, complexity, and observability of innovations. Results also showed that leveraging MDM and SOA as strategic resources enhances operational efficiency, fosters data quality, and strengthens decision-making. These findings illustrated the importance of unique, valuable, and inimitable resources in achieving competitive advantages, as posited by the Resource-based theory. The study additionally highlights the adaptability and reconfiguration of information technology (IT) systems to address market demands and consumer preferences. The integration of MDM and SOA aligned with the dynamic

Capabilities theory's emphasis on sensing, seizing, and transforming capabilities to maintain competitive advantages.

The research findings indicated that an effective data management framework positions a master data manager as a mediator between departmental operations and the enterprise database to ensure high-quality data storage and integration. This structure also includes data processing and enhancing operational and strategic functions. MDM and SOA focus on maintaining data quality, consistency, and decision-making across retail operations. Implementing MDM and SOA aids in collecting, storing, and analyzing data to support business insights and provide vital information on customer behavior and supply chain process and efficiency. Despite these benefits, challenges such as heterogeneous supplier data definitions, resource limitations, and integration blocks impede implementation. Addressing these obstacles is essential to leverage the potential of MDM and SOA for advancing organizational performance, customer satisfaction, sales, and employee productivity.

The findings also indicated customer data can be used in various systems during data lifecycle management in a supply chain planning system and Customer Relationship Management in a financial and ERP system. Also, the findings indicated implementing master data management and service-oriented architecture in the retail industry begins with engaging in Big Data optimization to use Big Data to improve retail operations. Big Data analysis may be conducted without a specific goal, and it is later determined how the insights can be used. Insights could also be real-time when the right software for real-time data analysis and insights is available to the retail company. The MDM and SOA architecture requirements need to be analyzed to enhance their efficiency. The findings

showed the main factors influencing data management model selection are technical experts, internal perception about MDM, previous projects related to managing master data, and internal culture. The results also indicated the desired characteristics of MDM architecture in the retail industry are availability and integration, architecture, governance, user interface, and lifecycle management. Implementing MDM and SOA architecture in the retail industry requires an architecture facilitates a more efficient way of managing master data in terms of data maintenance, data implementation, performing data modifications, and searching for specific data. Even though the management of master data is discussed for data within the same database, few companies are using one database to store all data. Therefore, the solution needs to manage data within the whole enterprise and not only within a system. To avoid the bottlenecks in integration that may happen when master data changes are performed, a user should preview the impact a change would lead to, thus decreasing the risk of conducting changes with unanticipated results in subsystems.

The results also revealed processes and rules must be established by the stakeholders involved to define details of how master data management should be done. Customer projects need large system configurations, and it is hard to learn about the different changes performed in such a configuration to identify the configuration possibilities. Companies would benefit from monitoring product data to eliminate irrelevant data and ensure no data is duplicated. Monitoring data implemented from integrated systems, particularly from companies with large suppliers, is vital to ensure agreements are kept about processes and products. Managing complex data phasing-out processes requires providing system support for product data management when the

product is in stock when it should no longer be purchased, and when it can be removed from the systems. The results also show SOA and MDM development should focus on understanding processes and services. Service design is one of the most challenging parts of MDM and SOA development. The outcome and process of implementing MDM and SOA depend on existing data architecture and systems. The compatibility of MDM and SOA software solutions and existing systems must be compared and evaluated to determine whether existing systems have been altered or the targeted MDM and SOA software to get interoperability.

The results also indicated data governance and security play a significant role in implementing MDM and SOA. Successful data governance results in the quality of master data, which is complete, timely, and reliable. Data security in data governance defines the data storage process, accessibility, and entry point to ensure the retention and privacy of master data. Integrated data enables an organization to use master data for decision-making and establishing a successful future for organizations. Data governance naturally impacts the company's strategy and influences business processes because requirements and rules limit the speed and flexibility of a process. However, technical and business experts may face the challenge of joining, managing, and controlling all data sources distributed over various business units. The research participants recommended managing master data based on defined standards, including integrity and data security processes. They also recommended observing and documenting the ongoing individual implementation of MDM in an organization to name tasks and responsibilities effectively.

Chapter 5

Conclusions and Recommendations

This study evaluated the role of effective data management systems and a robust event-driven architecture in measuring and acquiring business insights, such as increased sales, employee productivity, and customer satisfaction. This chapter presents the conclusions and recommendations drawn from the findings of this study. This chapter was essential because it synthesizes the results discussed in previous chapters and reflects on the broader implications of the research. By providing conclusions based on the research questions, the chapter offers actionable insights to guide future practice and research in the field. Additionally, this chapter highlights the study's limitations, providing a balanced understanding of its scope and constraints.

The chapter is structured as follows: first, the conclusions from the study are presented, summarizing the key findings and implications. Next, practical recommendations for retail industry leaders are outlined, aimed at improving the use of data management systems and event-driven architecture. Finally, the chapter concludes with suggestions for future research to investigate unresolved issues and build upon the current study's findings.

Research Questions

The research questions posed in this study were designed to explore key challenges and opportunities in using data management systems and event-driven architecture within the retail industry. The questions seek to understand the limitations of current systems, the impacts of Master Data Management (MDM) and Service-Oriented Architecture (SOA) on data quality, and the effectiveness of these systems in generating

valuable business insights. This section analyzed how the research questions align with the findings and how they inform the study's conclusions and recommendations for both industry leaders and future research efforts. The research questions below helped to determine the current challenges and limitations in using data management systems and event-driven architecture for measuring and acquiring business insights in the retail industry, the impacts of MDM and SOA on the quality and synergy of data across enterprises for the customer domain within the retail industry, and the impacts of MDM and SOA on the effectiveness of enterprise-wide data solutions in data quality, integration, and business insights in the retail industry?

Discussion of Findings

One of the aims of this research was to determine how implementing a data management system and sound event-driven architecture affects and measures business insights such as increased sales, employee productivity, and customer satisfaction. The findings in this study proposed that positive types of customer information, such as knowledge enhancing and action-oriented uses, would result in better customer performance. The symbolic application of customer information would result in the opposite outcome. The research established that action-oriented use of customer information improves customer performance. Knowledge-enhancing use of customer information harms customer performance. Knowledge-enhancing use of customer information effectively predicts a new product's performance, creativity, and timelines, while action-oriented use does not give insight into creativity in the development of a new product.

This research also established that an action-oriented user of customer information is also a determinant of the information used in a company. There can be various mediating factors between customer performance and knowledge-enhancing use of customer information. The factors could be the selection of a strategy and success in developing a new product. Knowledge-enhancing use of customer information can be hard to recognize by users because it does not entail directly applying the information to the issue being resolved. Therefore, it may not be perceived as information usage like action-oriented customer information.

Data Governance

The research determined data governance stands as the essential foundation for achieving MDM and SOA success. The research participants emphasized that the establishment of clear rules, roles, and policies leads to consistent data quality outcomes. Previous research proves that solid governance systems create better accuracy in data and enhances compliance while increasing decision-making speed. Data governance has been proven to reduce duplicate information according to studies that have already been published.

Master Data Management Frameworks

The research findings showed that a good data management framework includes placing a master data manager between an individual department and a general database.

These findings align with the findings of past studies that have explored data management frameworks. For instance, according to Khatri & Brown (2010), master data managers play a vital role in the data management framework by ensuring that the policies used on data governance are implemented properly. The authors add that data

governance therefore consists of the processes of data availability, data usability, data integrity and data security. An adequate framework in data governance leads to improved quality of data and operation by engaging the master data manager to avoid inconsistencies and unreliability of the data fed into the organization (Khatri & Brown, 2010). The data can be used and/or interpreted differently in different departments, resulting in diverse outcomes.

Master data is responsible for defining data standard and its format within the organization and is useful for ensuring data consistency and eliminating discrepancies that often cause problems in operations or result in misguided decisions. According to Otto et al. (2019), data consistency is an essential factor for the decision-making process and operations. Typically, data resides in functional organizational units or is stored on specific business units' servers, which makes it difficult to have a broad vision of an organization's data. The master data manager participates in data integration whereby the manager eliminates the barriers between departments and the common central base. This integration helps in improving the accessibility of data and to analyze data thus helping in improving performance. Integrated management of data also helps in getting insights and encouraging the decision-making process (Silvola et al., 2017). Another important aspect that the master data manager must consider is that data management must be treated as a scalable system, where the growth of the data volumes does not have to hurt the quality and access to it. Laney (2020) emphasizes the need to incorporate scalability into the current data management frameworks to accommodate the capabilities of modern-day organizations.

Implementing an Event-Driven Architecture

The findings of this study also indicated that implementing effective data management systems and a sound event-driven architecture within the retail industry helps to effectively measure business insights by providing vital information on customer behavior and the supply chain process. Therefore, electronic data is a significant component in retails which is why data management systems are important assets used in storage, acquisition, and interpretation of data. Such systems are effective in maintaining the integrity, usability, and security of data and, therefore, facilitate formation of corporate strategies from data collected by retailers. For example, marketing technology platforms enable retailers to consolidate and analyze customer information from customer interactions across the point of sale, digital, and contact centers. Thus, the accumulation of this information will allow retailers to develop an overall understanding of the customers' likes, frequencies of purchases, and actions.

According to Nguyen et al. (2018), proper data management means better customer categorization and targeted advertising and, therefore, higher levels of customer satisfaction and loyalty. There is also a need to achieve proper management of data supporting the supply chain management solutions by ensuring that suppliers, logistics, and inventory are integrated so retailers can easily manage the supply chain. Integrating all stakeholders in the supply chain aids in anticipating the demand rate, avoiding situations whereby one is left with no stock, and avoiding having large stocks that are unused. Chae (2019) pointed out that supply chain management using big data increases operations performance and decreases costs, thus increasing profitability.

Event-driven architecture (EDA) is another design paradigm of software systems characterized by a sequence of program events, such as users' actions, sensor readings, or messages coming from other applications. Although EDA is helpful when analyzing data in the retail business, this technique is applicable in the processing of real-time data. It also creates opportunities for retailers to reply to the actions of the customers in real time. This aligned with the Dynamics Capabilities Theory, which emphasized the ability of organizations to sense opportunities, seize them, and transform processes accordingly. Data analysis revealed that retail organizations capable of dynamically reconfiguring their MDM and SOA systems demonstrated superior adaptability to market demands. Participants provided examples of how their enterprises rapidly adjusted to shifts in customer behavior and market trends, showcasing the agility enabled by these technologies. This adaptability highlights the value of dynamic capabilities in navigating complex and fast-changing environments. For instance, when a customer orders a particular product online, an event is created, which includes changing the product's stock level, informing the warehouse, and sending the purchasing confirmation to the customer. This gives an immediate response to the customer and at the same time increases efficiency to the organization. Chen & Zhang (2020) noted that real-time processing of conversations enhances customer experiences since services are more friendly, personalized, and timely thus boosting customers' engagement and sales.

Retailers can use EDA to get real-time supply chain visibility and thus be able to address some changes in the supply chain, such as delays. For example, where a shipment is delayed, notifications of the delay and necessary action because of the delay are addressed to the required parties. This agility helps to establish better operations and meet

consumers' demands. Kamble et al. (2020) argued that real-time supply chain visibility through EDA also helps improve resilience and responsiveness as required in the management of risks associated with uncertainties in the retail industry. Integrating effective data administration systems with EDA helps retailers get more efficient and accurate measures of business intelligence. This joint delivery of the two perspectives allows having a more valuable and complete picture of the supply chain on the client side on one hand and client interaction and relevant processes on the other. Customer behavior analytics can assist in identifying the customers' preferences, furthering the capability of predicting future buying trends and ensuring corrective strategic marketing is in place. Information management and EDA also help monitor other essential supply chain effectiveness indicators like order delivery rates, time taken between ordering and receipt of goods, and frequency with which inventory is replenished. Metrics are monitored in real-time data and event processing so the retailers can assess the problem areas. Zhou et al. (2019) argue that the use of big data in managing the supply chain improves the results of decision-making and the organization's performance.

User Interface Usability

The users repeatedly noted usability issues with MDM systems as their main frustration. The acceptance of MDM systems depended on having a friendly interface, according to participants. Researchers have established that UI/UX design directly determines how users utilize systems while affecting their productivity level (Kamble et al., 2020). Results from IT adoption research support that complex interfaces generate user resistance, thereby endorsing the study findings.

Synergy of Data When Using MDM and SOA

The information gathered from participants in this study leads to the conclusion that implementing MDM and SOA improves the quality and synergy of data across enterprises within the retail industry for the customer domain from the perspective of an IT technician and business analyst by ensuring that accurate data is collected and analyzed appropriately. MDM is therefore the solution or management process used to handle several essential facets of data in an organization leading to accurate data collection accompanied by data integration. It also contributes to data accuracy, consistency, and reliability. In an IT technician's view, the company's MDM standard is important to maintain accurate and updated information. MDM ensures that there is reduction in data doublers since it provides an area where master data is stored. According to Otto & Schmidt (2017), MDM eliminates data inconsistencies hence improving the overall data quality for decision-making. MDM also strengthens data management in organizations by proving standard methods to be adopted by organizations in the ownership of their data. These include data cleansing, verification, and data monitoring to ensure that various rules relating to data management are being followed. For a business analyst this implies that he or she must be provided with adequate data that is credible, relevant and meets the set standards. According to Silvola et al. (2017), MDM helps in enhancing the quality and compliance of data due to strong data governance hence making data analysis and reporting reliable.

SOA is a particular architectural pattern that can allow services to interconnect over the network that will lead to the changeable and scalable interconnection of applications and data sources. SOA offers a means of creating and utilizing highly

integrated applications and data. It also plays the role of allowing data to be exchanged between different systems, thus ensuring that the customer data is well integrated.

According to Pulier & Taylor (2019), SOA can transmit, capture, and integrate real-time data, improving the efficiency of data management. This has enabled SOA to include the scale and flexibility aspects in the management of data within organizations. It also implies that new services and applications can be integrated smoothly without disturbing the existing frameworks. For a business analyst, it means the potential for involving new sources of data and analytical tools, which increases the ability to solve customer data analysis. According to Zimmermann et al., 2020, SOA is effective in scale and versatility, which allows organizations to address the fluctuations in terms of requirements on businesses as well as development in the technologies that are applied.

MDM stands for Master Data Management which is a place where master data resides and is owned. SOA is a means of getting master data to multiple applications in an organization. This makes it possible for all the systems to acquire correct and dependable data from other systems in the organizational environment. From the IT technician's perspective, it raises the overall mechanization of data handling and enhances the quality of data. The need for enhancing the quality and homogeneity of data in the enterprise when implementing MDM and SOA is in a study by Sen & Sinha (2018). To a business analyst, SOA and MDM interact to enhance the performance of data analysis. Thus, with the help of accurate and normalized data and the ability of SOA to integrate in flexible ways, analysts can apply more detailed analytical tools to gain a more profound outlook on customers. Such cohesive methodological identification is useful in business decision-making and the creation of efficient advertising

conceptualization plans. Verhoef et al. (2016) argued that data analysis is advanced to optimize customer information and business outcomes through MDM and SOA integration.

Challenges to MDM and SOA Implementation

The results also indicated that the main challenges encountered in implementing MDM and SOA are difficulty reading and inserting data in the IFS database because different suppliers have various definitions of a product. Regarding the implementation of MDM and SOA, one of the major issues that one can experience is dealing with the inconsistencies of product definitions that suppliers can provide. The products can be labelled differently, the structure of these products can vary from supplier to supplier, and even the basic schema of approaching data storage might not be the same. According to Otto & Schmidt (2017), differences in the definition of products cause tremendous problems in data integration and management. Standardized product definition is also not maintained well and thus gives rise to data quality problems like replicated data, missing information, and wrong information. In the case of an IT technician, data quality entails a lot of data processing as cleansing and transformation, hence making it very costly. Silvola et al. (2017) identified that the poor quality of data is the key issue that hinders the effective implementation of MDM which can create a need for constant updating and tweaking.

Combining data from multiple suppliers into the IFS database is intricate due to the added data heterogeneity. Optimization integration can be easily achieved using SOA since it is an adaptive integration architecture (Pulier & Taylor, 2019). However, the creation of the architecture and the constant maintenance are a complex process. Pulier &

Taylor (2019) noted that problems with data integration can delay MDM and SOA projects and the general market functionality. Another concern is that multiple suppliers reduced the efficiency of data sharing since the product data definition could be set differently across the suppliers. Standardization enhances the accuracy and soundness of data to be used in the analysis. According to Zimmermann et al. (2020), compliance with these standards can facilitate efficiency for organizations and suppliers when collecting and entering data into the supply chain. Sen et al. (2018) established that proper data governance enables the data to be accurate, complete, and consistent hence enhancing the integration and usage of data in the enterprise. There are various middleware and ETL (Extract, Transform, Load) tools that can help to facilitate achieving common data integration. These tools can be used to parse and purge data before loading it into the IFS database. To an IT technician, these tools eliminate any chance of disruption and boost the speed of data integration operations. According to Crowe et al. (2011), it is important to state that the conception and implementation of the ETL function in large-scale organizations is characteristically complex and is often executed using professionally developed tools.

Understanding the Data Lifecycle

This study also revealed that customer data can be used in several systems in the data lifecycle in the SC planning system and Customer Relationship Management in the financial and ERP systems. Data lifecycle management (DLM) refers to the management of data right from the creation stage up to its use and eventual disposal as irrelevant and deleted. When planning the supply chain, DLM is expected to handle and use customer information in the most efficient manner for the supply chain processes. The first step

followed in DLM is collecting information from the customers through inquiries, selling transactions, or feedback from the customers. Such information gathered from these sources is taken to the supply chain planning system that offers a customer's requirements overview. Hofmann & Rüsç (2017) support this assertion, stating that if customers' data are integrated into the supply chain system, the probability of predicting the demand is improved, and thus, inventory was well managed and stockouts minimized. The data extracted from customers was analyzed to establish trends and patterns of their buying behavior. Advanced analytics tools, machine learning algorithms, and forecasts are utilized from past information to come up with future demand. This predictive analysis, in return, enables supply chain managers to make very informed decisions about production planning, inventory management, and procurement. A study by Min (2019) proved the significance of predictive analytics in enhancing supply chain responsiveness and agility. The author argues that customer data analysis supports strategic decisions in supply chain planning. For example, insight into customer preference and demand patterns will let companies manage production and distribution strategies optimally. From an operational point of view, it gives better efficiency and lowered costs. Data-driven decision-making in SCM also improves overall performance and raises customer satisfaction.

Lifecycle Management

The research study participants showed whole-life data management solutions as essential for handling data before and after its creation phase. Such an approach matches recommendations in literature that support lifecycle-oriented architectures to boost data and reduce their expiration rates. Dynamic Capabilities Theory validates this discovery

because lifecycle management allows organizations to both detect and reply to changes in effective ways.

Customer Relationships Management Systems

Customer Relationship Management (CRM) systems are aimed at managing the company's interaction with its current and potential customers. CRM is in the financial and ERP systems because it gives a holistic view of customer data, hence enabling the company to serve customers better and strategize (Ngai et al.,2020). Customer data can be integrated into CRM systems by consolidating information from various touchpoints, such as sales, marketing, and customer service. The integration ensures that all departments always work with similar customer information. Ngai et al. (2020) stated that proper integration of the CRM system with the ERP system results in improved accuracy and consistency of data and provides better insight into the customers' systems also analyze customer data to gain insight into the customers' behavior, preferences, and needs. These insights are used to adapt any marketing campaigns, improve customer service, and develop customized products. From a financial perspective, information about customer behavior proves helpful in the identification of high-value customers and the optimization of pricing strategies.

Big Data Optimization

The findings of this study also indicated that implementing master data management and service-oriented architecture in the retail industry begins with engaging in Big Data optimization to use Big Data to improve retail operations. Big Data Optimization is utilizing the key volumes of data emanating from different sources to derive relevant insights that improve the organizational business processes. Big Data is

structured and unstructured data from sources like customer transactions, social media interactions, and sensor data, among many others. Big Data, in the retail sector, can be optimized toward improving various dimensions of operations about customers, inventory management, and supply chain efficiency. Big Data optimization is usually applied right after Big Data that has been obtained from various sources is unified and a single point of reference is created (Silvola et al., 2017). This forms the base on which MDM stands since it is informed by consistency and accuracy. Big Data technologies can be used to collect data from ‘‘point of sale’’ systems, e-shoppers, and consumers’ feedback in one central database. As pointed out by Silvola et al. (2017), the quality and consistency of the data are key to the success of the Data integration activities that precede it, which allow MDM to be achieved successfully.

Big Data optimization eliminates repeats, mistakes, and errors that are usually present in such data. Thus, high-quality data is essential for MDM because it enables the formation of the master database, a tool that is credible and devoid of significant error. It is, therefore, possible to implement the process of data cleansing through Big Data analytics using advanced data cleaning strategies, which automate the process, hence reducing the time that would be used for manual operations. A study by Wang et al. (2019) highlights the importance of data quality management in MDM, regarding which an attempt has been made to describe how data can be made more accurate and compliant through automated data cleansing. Integration of MDM provokes norms of data management to protect data, monitor the aspects of privacy, and ensure data requirements are met. Aspects of governance about policies touching on the particulars of big data technologies and the detection of data usage in organizations, are some advantages that

come with its use. This ensures that master data is managed as required by the legal requirements and the organizational guidelines.

According to Otto & Schmidt (2017), data governance helps with data integrity and compliance, which supports MDM success. Servicing integration and interconnection is one of the steps in the realization process when implementing SOA. Big Data optimization also means that complex systems and applications that interact and exchange data with each other in the business are intertwined (Otto & Schmidt, 2017). This enables retailers to harness Big Data technologies to enhance the establishment of a service-oriented architecture for real-time interactions in data sharing. Khodakaram et al. (2018) argue that Big Data integration in an SOA setting can enhance the inter-operation of the services within the framework and enhance the operations' operational effectiveness.

Big Data optimization promotes scalability since it deals with the processing and analyzing of large data in real-time. This, therefore, will enable a retailer to respond quickly to changes and changes in customers' preferences. Another enabler of SOA within e-commerce includes the scalability of Big Data technologies because it promotes flexibility to adapt to ever-shifting business needs. Optimized Big Data implemented in real-time can be useful for retailers to learn several things about their clients, their sales, and the performance of their business. These facts can be incorporated into the SOA framework to achieve an analysis of trends to improve the responsiveness and flexibility of retailing. Hofmann and Rüsç (2017) pointed out real-time analytics in SOA as a facilitator of proactive decision-making capacity supplementing business performance in an organization.

Big Data optimization also gives retailers the chance to know details about the customers' tastes or changes in behaviors. Such information may be used for the personalization of advertising campaigns, upgrading products, and increasing the level of service to customers (Hofmann & Rüsç,2017). Thus, incorporating these ideas into an MDM and SOA framework guarantees all forms of communication with customers to be personable and coherent, hence bolstering customer satisfaction and loyalty. Hofmann & Rüsç,2017) further argue that the capability to generate value from big data in developing customer experience is one of the biggest advantages that one retailer has over the other. Big data optimization also facilitates operational fluidity in the retailing process through timely and accurate information. Implementing MDM also facilitates the definition of data in all the systems while eliminating errors and duplication. This also ensures that service-to-service data exchanges are efficient because SOA comes with an architectural design that is easy to implement (Hofmann & Rüsç,2017). The usage of MDM and SOA enabled through data implementation aids in enhancing operation capacity and reducing cost.

Big Data can be optimized to enable retailers to improve their demand forecast accuracy and inventory level management. Integrating this information into an MDM and SOA framework makes inventory data across all systems uniform and consistent. Thus, the risk of stock-outs and overstocking will decrease, resulting in better inventory turns and lower carrying costs (Min, 2019). Thus, data-driven inventory management is important in ensuring that optimal stock levels and customer demand are observed (Min, 2019). However, it is difficult to integrate data from different sources. Retailers should, therefore, utilize advanced Big Data technologies and tools that can permit the simplicity

of data integration. According to Crowe et al. (2019), effective integration of data facilitates complete Big Data optimization regarding MDM and SOA. Great volumes of customer data also create concerns regarding data security and privacy. The retailer must support this with robust security measures such as encryption and access controls to ensure the safety of sensitive information. In data protection, compliance with GDPR or similar laws becomes very important. According to Crowe et al. (2019), setting up protection for data security and privacy is a step that gains customer trust and prevents customers from probable legal action.

Scaling Big Data solutions to high volumes of data and processing requirements is a potential challenge. To grow with the retailer's needs, Big Data systems require an investment in infrastructure and solutions that are scalable, and cloud based. The independent implementation of distributed computing frameworks, including Hadoop and Spark, could enhance scalability and performance (Gandomi & Haider, 2017). Gandomi and Haider (2017) confirmed the need for Big Data Infrastructure in the implementation of MDM and SOA. Big data and advanced business intelligence, along with artificial intelligence and machine learning, are slowly but steadily gaining their foothold in Big Data and have a tremendous scope to further improve MDM and SOA implementation. AI and ML could be more precise and useful in making conclusions regarding the analyzed data due to process automation in data integration, cleaning, and analysis. According to Verhoef et al. (2016), Big data, with AI, offers itself as a means through which the capacity for decision-making and functioning of retail organizations could be enhanced. Employing blockchain is essential to developing secure and relatively open methods for handling data in distributed systems. Blockchain can bring changes in data

management and MDM and SOA frameworks with the help of its sensing capability for recording the transactions done on data accurately and without duplication.

Considerations in Choosing the Data Management Model

The results also confirmed that the possibilities of choosing the data management model depend upon technical specialists, the perception of MDM by the company's employees, former projects about managing master data, and company culture. There is a high connection between the presence and number of technical experts in an organization and the choice of the model for data management. There is also a great need to get specialists to put in place forms of data management systems like master data management and MDM. Silvola et al. (2017) recognized that the central enabler of MDM includes having technical experts with long-term experience in managing the data and with specialized knowledge of data management technologies and the best practices for their application. Organizations that prioritize training and development can adopt the described models of managing data. Organized continuous learning and professional development ensure the technical staff is familiar with the latest trends and technologies. A study of the challenges facing data management and use by Gandomi and Haider (2017) asserted that investment in training by the technical staff was crucial. Organizational factors that influence the choice of data management models are those that consider the collaborative approach of technical experts within a certain organization in sharing knowledge. Organizational factors in the selection of models for data management are inclusive of the collaborative environments where the experts gather and share their ideas and best practices, hence making better-informed decisions about models of data management. According to Otto & Schmidt (2017), the culture of

collaboration among technical experts helps the organization better implement and gain sustainability in proper data management practices. Internal perception of MDM is critical in determining the choice and ultimate model for data management. Knowledge of the advantages and disadvantages of MDM informs the decision-making process. A survey by Silvola et al. (2017) found that an increased awareness of MDM's strategic advantages ensured that the organization committed to adequate data management models.

The highly efficient way to instill good internal views on MDM would be through leadership support since the individuals will guide the organization on treating data as the strategic asset of the organization through their leadership on data programs and clear communication on the value that MDM brings with it (Smith & McKeen, 2017). Smith and McKeen (2017) added that the organized readiness of data management projects and the fit of the data management adoption model are directly impacted by leadership support to MDM efforts. Therefore, internal subjectivity on data management impacts organizational readiness in an organization to embrace this initiative. Organizations that consider data a critical asset will be likely to invest more in stronger data management environments. Research work by Khatri & Brown (2010) highlighted the significance of assessing organizational readiness and aligning the same with the MDM strategies to ascertain an effective implementation process. Previous master data management projects provide a rich source of information that influences the selection of the data management model. In the experiences derived from successes and failures, the organization can establish what works and vice versa. According to Otto & Schmidt (2017), experiences made from past projects are considered to help in making informed decisions about the

selection of the data management model. Organizations that have had data projects done in the past are also more likely to have some form of infrastructure and processes that can be harnessed in building up new projects. The building of new capabilities may ensure an easier transition and a much faster implementation of new models regarding managing data. This also follows the paper by Liu et al. (2018) which cites the use of existing infrastructures in augmenting efficiency and effectiveness among data management projects. According to Silvola et al. (2017), organizations that systematically evaluate the outcomes of conducted projects are also in a better position to make decisions related to the models of data management applied in the organization in a way that serves their goals.

An organization's internal culture, and particularly its value of data, registers high influence regarding the choice of a data management model. An organization with a strong data-driven culture regards data as an asset, and therefore, such an organization will adopt advanced data management models. High-level data management models that increase data accessibility and quality gain more acceptance among organizations with a strong data-driven culture. Internal culture also influences how people within the organization interact and communicate with each other. A culture that encourages the free flow of ideas and working together cross functionally is in a better position when adopting information management models. A study by Silvola et al. (2018) revealed that organizations with collaborative cultures were more successful at conducting MDM initiatives and attaining the envisioned results.

Desired Characteristics of MDM Architecture

This study also established that the desired characteristics of MDM architecture in the retail industry are availability and integration architecture, governance user interface, and lifecycle management. From a retail business perspective, operations occur at an extraordinarily dynamic pace. Real-time synchronization ensures that all the systems are armed with the most recent data to make timely decisions (Wang et al.,2019). Wang et al. (2019) added that the integrated MDM architecture with real-time data synchronization increases operational efficiency and provides the capability to respond to the market. Some of the main characteristics of MDM architecture in the retail sector include scalability. Each increasing level of growth by a retail business amounts to an exponential increase in data volume. Scalable MDM architecture handles growing data volumes without affecting performance. Therefore, it ensures efficiency and responsiveness in its operation while the business is growing. According to Gandomi and Haider (2017), scalable MDM architecture supports the long-term growth of a business. This is in terms of its specified data management needs. Flexibility in MDM architecture allows retailers to stay updated on the changing business requirements and market conditions. In flexible MDM systems, with new data sources, business processes are managed effectively.

The results also show that stakeholders must set up processes and rules to define the details of how master data management should be done. If one considers data validation, cleaning, and enrichment, proper processes and rules should be in place to ensure good data maintenance. In organizations with good quality management processes in place, master data can easily be retained as accurate and reliable for decision-making and operational efficiency. The development of the MDM processes and rules should

support the abilities of relevant stakeholders to ensure that the guidelines are comprehensive enough to meet the needs of the whole organization. In the development of the process and rules, consideration of diverse requirements and organizational perspectives can be ensured through stakeholder involvement at different levels in IT, finance, marketing, and operations departments.

A well-defined role and responsibility framework is among the crucial things in developing processes and rules for MDM, whether the stewardship of data arises from data stewards, data owners, or other key individuals responsible for the management of master data. Gandomi and Haider (2017) emphasized role definition regarding MDM by arguing that clarity of roles and responsibilities instigates accountability, hence the smooth running of activities within data management. The biggest challenge connected to MDM is the high quality of data. An assessment and continuous monitoring of the quality of data is crucial as it provides resources to identify troubling issues that can affect the data quality. According to Gandomi & Haider (2017) organizations that have implemented the most current technology of data quality management are likely to maintain master data accuracy and reliability. The findings show that the principle of design for SOA and MDM development should be recognized in processes and services.

Challenges in Service Design

The process of service design proves to be one of the most challenging in the development of MDM and SOA. Service design proves to be detailed and complicated in both developments of SOA and MDM. It entails recognition of the functionalities that the services would offer, how they interrelate, and how they connect with other services or the entire systems. For service design to be effective, a good understanding of the

business process and data requirements is important. For effective design, Ibrahim et al. (2020) mentioned that the design of functional but flexible services is well planned and clarified regarding the business objectives and technical constraints that would be involved. One of the biggest challenges in service design is flexibility and reusability. Services must be architected in a manner that they also become reusable across applications and processes besides being flexible examples enough to accommodate dynamic changing business needs. Taran et al. (2019) state this requires that services are designed with clear interfaces and services that could handle multiple scenarios without significantly changing their configurations. Another major challenge to service design is interoperability between services. SOA requires services to communicate and interact with each other flawlessly. In a study carried out by Silvola et al. (2018), it was concluded that standardizing the protocols of communication and data formats eased interoperability and integration among services. Thus, to develop a service properly, the business process needs to be understood. There is also a need for such processes to be mapped, key data elements identified, and the definition of the interactions between the services achieved (Papageorgiou et al., 2018).

Incorporating the business process with the design of the service can be useful as it can make the services align with the needs of the business and, at the same time, support operations making it efficient. According to Papageorgiou et al. (2018), process modeling and analysis turn out to be twin activities in designing services to better the performance and integration of business processes. This can ensure that services are defined to focus on specific functionality. This describes what each service does, what inputs it shall take, what the outputs shall be, and what it will communicate with other

services. According to Liu et al. (2018), a clear boundary definition among services avoids the overlapping of services and redundancies, ensuring there is an efficient and well-manageable service architecture. According to Taran et al. (2019), best service design practices can be adapted to meet the several challenges that service design faces. They include service structuring, which is effectively done through the design patterns of the Service Layer pattern and Composite Service pattern, which propose tried approaches to addressing common challenges that occur in service design.

SOA and Data Security

SOA helps MDM in integrating several data sources and systems. It provides a framework for creating interoperable services that can communicate with different data sources and applications. According to Silvola et al. (2018), data governance in SOA concerns service interface management, data contracts, and service interactions for interoperability and compliance with service standards. Effective data governance in SOA requires maintaining data security. Data security could be called the process that ensures data protection from unwanted accesses, breaches, and several other sorts of threats. This would relate to encrypting data, controlling access, and masking it.

According to Reddy and Gupta (2018), data security is needed in safeguarding sensitive information and compliance with regulations in data protection, which often becomes very critical for successfully implementing MDM and SOA. Security for data in MDM is all about the protection of master data from unauthorized access and breaches (Gandomi & Haider, 2017). According to Gandomi & Haider (2017), ensuring data security in MDM includes setting access controls through role-based access control models, mastering data based on the roles of the users, protecting sensitive data both in

rest and transit using encryption techniques to maintain security so the data remains safe from any unauthorized access and applying data masking techniques on sensitive information in non-production environments. This will minimize the risk of data exposure since there could be the possibility of some users accessing these non-production data stores who do not need to know.

Security in SOA is essential to ensure the protection of data exchanged between services, and likewise, the operations of services should be secured. Security techniques in SOA are authentication, authorization, and secure protocols of communication.

According to Zhang et al. (2020), data security in SOA includes developing robust authentication and authorization methods that give access and allow interaction with the system only to legitimate users and services, promoting secure communication, and defining and enforcing security policy for services, including access controls and data protection measures, audit logging. Here, data governance and security frameworks need to be integrated into MDM and SOA. According to Verhoef et al. (2021), both governance and security frameworks have their goals and practices oriented to complement the application of data management and security provisions supporting the total objectives set by MDM and SOA. The major strategies on integration include developing a unified approach to policies and procedures that address data governance and data security, ensuring consistency of this across MDM and SOA, collaboration among data governance and security teams on common problems, and aligning strategy toward managing and protecting data and monitoring and continuous improvement to review and perfect the governance and security practices regarding developing risks and requirements.

Limitations

The researcher experienced various limitations when conducting the study. One of the limitations was the sample size. The number of participants that completed this study was twenty participants (minimum) who were self-selected. This sample size was small, thus limiting the ability to generalize the research findings. The second limitation was bias. The researcher relied on information provided by self-selected research participants, which could increase the risk of bias because the research participants may not have accurately represented the broader population. The third limitation was time constraints. The researcher experienced time constraints because most research participants did not respond to the request to schedule an interview.

Recommendations to Leaders and Practitioners

Modern technology is taking over most processes within the retail industry. This study has established that Master Data Management and Service-Oriented Architecture can improve efficiency in the supply chain in the retail industry by enabling retail companies to efficiently manage the flow of goods across the supply chain and anticipate the demand for specific products based on customer preferences. Therefore, I would make four recommendations. First, implement better governance frameworks. Essentially, retail leaders should train employees on how to use Service-Oriented Architecture and Master Data Management to address the challenges in the supply chain and other processes in the retail industry. Second, I recommend prioritization of usability in system design. Retail leaders should involve end-users during UI/UX development to ensure the MDM interface supports productivity. Design simplicity encourages higher system adoption. Third, businesses should invest fully in lifecycle data management. The

systems adopted should support data from creation to retirement. Notably, this approach aligns with the dynamic needs of supply chains. Lastly, I recommend that business should conduct architecture readiness assessments. Essentially, leaders should evaluate their existing IT infrastructure for integration compatibility before adopting MDM and SOA. The deployment of legacy systems through phased migration plans helps lower both disruption levels and implementation danger points.

Recommendations for Future Research

Current research has addressed how the implementation of data management systems and sound event-driven architecture systems affect businesses and measure business insights such as increased sales, employee productivity, and customer satisfaction. The research has also determined ways integrating MDM and SOA in the retail sector improves the quality and synergy of data across organizations within the industry for the customer domain from the view of an IT technician and business analyst. Therefore, future studies should focus on the challenges encountered in the implementation of MDM and SOA in the retail sector and how retail companies are addressing the challenges. Future studies should also focus on the current trends in the use of MDM and SOA in the retail sector and how they are impacted by changes in the retail business environment.

Chapter Summary

This study focused on evaluating the role of effective data management systems and a robust event-driven architecture in measuring and acquiring business insights, such as increased sales, employee productivity, and customer satisfaction. One of the main findings of the study was that a good data management framework requires placing a

master data manager between an individual department and a general database. The second main finding was that implementing effective data management systems and a sound event-driven architecture within the retail industry helps to effectively measure business insights by providing vital information on customer behavior and the supply chain process. The third main finding was that implementing MDM and SOA improves the quality and synergy of data across enterprises within the retail industry for the customer domain from the perspective of an IT technician and business analyst by ensuring that accurate data is collected and analyzed appropriately. The fourth main finding was that the main challenges encountered in implementing MDM and SOA are difficulty reading and inserting data in the IFS database because different suppliers have various definitions of a product. The fifth main finding was that customer data can be used in various systems during data lifecycle management in a supply chain planning system and Customer Relationship Management in a financial and ERP system. The sixth main finding was that implementing master data management and service-oriented architecture in the retail industry begins with engaging in Big Data optimization to use Big Data to improve retail operations. The other main finding was that the main factors influencing data management model selection are technical experts, internal perception about MDM, previous projects related to managing master data, and internal culture.

The results indicated that the desired characteristics of MDM architecture in the retail industry are availability and integration, architecture, governance, user interface, and lifecycle management. The results also revealed that the stakeholders must establish processes and rules to define details of how master data management should be done. The results showed that SOA and MDM development should focus on understanding

processes and services. Service design is one of the most challenging parts of MDM and SOA development. The results finally indicated that data governance and security play a significant role in implementing MDM and SOA.

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Appendix A

Informed Consent Form



Informed Consent: Participants 30 years of age and older

Greetings,

My name is [REDACTED] and I am a doctorate student at the University of Phoenix in which I am earning my degree in Doctor of Management in Organizational Leadership. I am conducting a research study entitled Implementation of Master Data Management and Service-Oriented Architecture in the Retail Industry: Action Research.

The purpose of this research study is to review the impact of the implementation of a Master Data Management (MDM) and Service-Oriented Architecture (SOA) in the retail industry by measuring and acquiring business insights such as return on investments, employee productivity, and customer satisfaction.

Your participation will involve a recorded interview, which will include 12 open-ended questions related to the study. The expected time committed would be approximately 45- 60 minutes and will be conducted via ZOOM. The interview will be audio recorded for the researcher to reference when transcribing the information. At any time during the interview process, the researcher may ask you to elaborate in any of the interview questions that may not be clear. This may initiate further discussion that contribute to gathering as much information as possible.

You can decide to be a part of this study or not. Once you start, you can withdraw from the study at any time without any repercussions. The results of the research study may be published but your identity will remain confidential, and your name will not be made known to any outside parties. Informed consent forms using electronic signatures will ensure that you have agreed to the terms and understand their role in the study and the rights that they will be protected at all costs.

In this research, there are no foreseeable risks to you. Minimal risk to participants will be addressed by conducting the interviews online and maintaining confidentiality.

Although there may be no direct benefit to you, a possible benefit from your being part of this study is to contribute to the information technology field and retail business. Having a group of participants who can give their perspective on a retail business model and how information data systems can support or fail if implemented incorrectly can be extremely valuable.

If you have any questions about the research study, please call me at [REDACTED] or email me at [REDACTED]. For questions about your rights as a study participant, or any concerns or complaints, please contact the University of Phoenix Institutional Review Board at IRB@phoenix.edu.

As a participant in this study, you should understand the following:

1. You may decide not to be part of this study or you may want to withdraw from the study at any time. If you want to withdraw, please call me at [REDACTED] or email me at [REDACTED]. The data will be withdrawn from the study after the participant has been interviewed/surveyed. Recordings, transcriptions, and any notes taken during interviews will be deleted.
2. Your identity will be kept confidential.
3. [REDACTED] the researcher, has fully explained the nature of the research study and has answered all of your questions and concerns.

4. Before interviews are conducted, you must give permission for the researcher, [REDACTED] to audio record them. The information from these audio-recorded interviews will be transcribed by researcher, and the data will be coded to assure that your identity is protected.
5. Any/all applicable COVID-19 protocols/guidelines will be followed to minimize potential health risks to participants.
6. Your data will be kept secure and encrypted on a computer file. All participant data will be maintained by referring to you as a research participant or respondent, such as Participant 1, Participant 2, etc. rather than using your name. Any of your personally identifiable information (PII) and study data collected will be stored separately on an external encrypted drive and locked with passwords.
7. Audio-recordings will be destroyed once transcripts are reviewed and approved by participants. The approved transcripts are kept for three years. The data will be destroyed by deleting all computer files containing audio recordings and transcriptions. Paper study materials WILL NOT be kept.
8. The results of this study may be published.

By signing this form, you are confirming that you meet the following inclusion criteria for participation in this study.

1. Over the age of 16 years old and below the age of 62
2. Work in the field of Master Data Management and Retail Industry
3. 5+ years of technology implementation experience in the area of Master Data Management

By signing this form, you agree that you understand the nature of the study, the possible risks, and benefits to you as a participant, and how your identity will be kept confidential. When you sign this form, this means that you are 30 years old or older and that you give your permission to volunteer as a participant in the study that is described here.

() I accept the above terms. () I do not accept the above terms. (CHECK ONE)

Signature of the research participant _____ Date _____

Signature of the researcher _____ Date _____

Appendix B

Interview Questions

R1. What is the role of effective data management systems and a sound event-driven architecture in contributing to the process of measuring and acquiring business insights?

Sub questions:

1. How does the current state of data management impact business decisions?
2. What are the key attributes of successful data systems in retail?
3. How do data management systems facilitate predictive analytics and forecasting?

R2. What is the impact of MDM and SOA on creating process improvement in quality and synergy of data across enterprises within the retail industry for the customer domain?

Sub questions:

1. What challenges arise during the integration of MDM and SOA?
2. How do these systems address data silos and redundancy?
3. What strategies have been effective in scaling MDM and SOA solutions across diverse retail operations?

R3. What are the impacts of implementing effective enterprise-wide data solutions in the retail industry?

Sub questions:

1. How do IT professionals define best practices for data infrastructure?
2. What outcomes are achieved through enterprise-wide data solutions?
3. How do enterprise-wide data solutions contribute to customer satisfaction and loyalty?

AUTHOR BIOGRAPHY

Dr. Usen Victor Uboh, a Philadelphia native, is a lifelong learner and passionate advocate for effective leadership within the Information Technology industry. He earned a Bachelor of Arts in Management Information Technology from Temple University and a Master of Business Administration from the University of Phoenix. Throughout his academic and professional career, he has consistently shown a commitment to excellence, innovation, and the ability to lead diverse teams to achieve organizational objectives. With over 20 years of experience in Information Technology, specifically in Data Architecture, Usen has held pivotal roles, contributing to strategic planning, change management, and leadership development initiatives. The pursuit of a Doctor of Management in Organizational Leadership at the University of Phoenix reflects Dr. Usen's dedication to advancing knowledge and creating impactful solutions in the fields of leadership and organizational management. His dissertation research focuses on Master Data Management and its impact on the retail industry, aiming to contribute to both academic literature and practical leadership frameworks.

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